



On optimal fixed time stability: Its application in secure communication using sliding mode control

Rajesh Kushwah , Arnab Mapui *, Santwana Mukhopadhyay

Department of Mathematical Sciences, Indian institute of Technology (BHU), 221005, Varanasi, India

ARTICLE INFO

Keywords:

Finite time stability ($F_T S$)
 $FX_T S$
 SMC
 Chaotic/hyper chaotic systems
 Secure communication scheme (SCS)

ABSTRACT

One of the main contributions of the present article is to provide some sufficient conditions for fixed-time stability ($FX_T S$) of nonlinear dynamical systems and the superiority of the proposed conditions over several such conditions already available in the literature. In this respect, a theorem on $FX_T S$ is established, and an estimation of the settling time is provided. Furthermore, additional generalized forms are investigated to improve the application of the proposed results. The present results are further applied to a secure communication system to achieve fixed-time synchronization of master-slave system. In secure communication, to prevent data loss or distortion of the transmitted message, it is essential to effectively reject disturbances and uncertainties. In order to address this issue and achieve robust synchronization in the presence of channel noise, a sliding mode control scheme is developed in the present work. The proposed SMC not only guarantees that the synchronization error converges to zero in fixed time but also provides strong disturbance rejection capability. In this respect, some non-singular terminal sliding surfaces (NSTSS) and corresponding controllers are designed. At last, some numerical simulations are provided to verify the theoretical results.

1. Introduction

Stability is a crucial dynamic characteristic of nonlinear systems, and the research on it has received plenty of attention in recent years. Among the various stability concepts, asymptotic stability (Atassi & Khalil, 2000; Hale & Collection, 1980; Hale & Lunel, 2003; Khalil, 2002) and exponential stability (Cao & Wang, 2002; Istratov & Vyvenko, 1999) have been investigated intensively. Asymptotic stability and exponential stability are categorized under infinite-time stability, where the system trajectories converge to desired state as time tends to infinity. This indeterminacy poses limitations in practical systems such as power systems, robotic manipulation, missile guidance systems, etc., where rapid convergence and enhanced response performance are required. To address this drawback of infinite stability, Bhat and Bernstein introduced the notion of finite-time stability ($F_T S$) to the control community (Bhat & Bernstein, 2000) in 2000. In $F_T S$, trajectories of system converge to the equilibrium point within a finite time for any initial condition. This convergence time is referred as the settling time. During the past several years, $F_T S$ has been given much attention to get optimal convergence time and robustness against uncertainties (Haddad et al., 2008; He et al., 2021; Moulay & Perruquetti, 2006, 2008; Xiao et al., 2021; Yu et al., 2005). Though the notion of $F_T S$ has an

upper hand over several notions of infinite-time stability, it suffers from the drawback of dependency of the settling-time on the initial condition of the system. Consequently, for different values of initial conditions, settling time is found to be different. Also, the initial conditions of many practical system is hard to determine or impossible to know in advance. To overcome this drawback, the concept of $FX_T S$ was introduced by Polyakov (2011). $FX_T S$ offers the advantage of the existence of a uniform bound of the settling time, regardless of the initial conditions. Further, the obtained upper bound of the settling time is constant that depends only on the system's parameters and is independent of the initial conditions. Polyakov (2011) first derived a Lyapunov inequality $\dot{V}(X) \leq -(b_0(V(X)))^{\frac{1}{\mu}} + c_0(V(X))^{\frac{1}{\nu}}$; $X \in \mathbb{R}^n \setminus \{0\}$, $b_0, c_0, k > 0$, $0 < \mu < 1 < \nu$ for $FX_T S$ and estimated the upper bound of the settling time. A few years later, Hu et al. (2017) revisited the formulation presented in the article (Polyakov, 2011) and provided a more accurate estimate of the upper bound of the settling time. Subsequently, an alternative Lyapunov inequality incorporating a positive linear term was proposed in Li et al. (2019) under suitable parameter conditions. One year later, Hu et al. (2020) investigated a more general form, where linear term could be positive, negative or zero, which extended the work of Li et al. (2019). Further, several new Lyapunov inequalities using the signum function are reported in the work (Kong & Zhu, 2021; Li et al., 2020;

* Corresponding author.

E-mail addresses: rajeshkushwah.rs.mat23@itbhu.ac.in (R. Kushwah), arnabmapui.rs.mat20@itbhu.ac.in (A. Mapui), mukhosant.apm@iitbhu.ac.in (S. Mukhopadhyay).

<https://doi.org/10.1016/j.ejcon.2026.101467>

Received 10 September 2025; Received in revised form 31 December 2025; Accepted 30 January 2026

Available online 3 February 2026

0947-3580/© 2026 European Control Association. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Mapui et al., 2025; Wang et al., 2024; Xu et al., 2023; Zhang et al., 2024) for FX_T -S analysis. So, over the years, researchers have focused on enhancing the sufficient conditions to obtain FX_T -S of corresponding systems. Obtaining an enhanced sufficient condition for achieving FX_T -S is one of the goal of the present article.

Furthermore, to achieve the desired system behavior, suitable control inputs are required. Recently, extensive research has been devoted to the stability analysis of nonlinear systems using various control techniques, such as adaptive control (Chen & Lu, 2007), backstepping control (Vaidyanathan & Azar, 2021), PID control (Launay et al., 2006), SMC (Afghoul et al., 2018; Levant, 2013; Utkin et al., 2020; Utkin, 2013), among others (Ahmad et al., 2015; Fu et al., 2021; Sun et al., 2020). Among all of these control techniques, SMC has become very popular because of its simplicity and robustness against uncertainty and external disturbance. SMC consists of two parts: first, a switching manifold (surface) is constructed to define the desired convergence rate, and then a controller is developed to force and maintain system states on this manifold. In classical SMC, the switching manifold is a linear hyperplane, indicating that systems state trajectories approach the origin asymptotically. To obtain finite time convergence of state trajectories, terminal sliding mode control (TSMC) (Babes et al., 2019; Mu & He, 2017; Wu et al., 1998; Zhihong et al., 1994) and fast terminal sliding mode control (FTSMC) (Balta et al., 2023; Pan et al., 2020; Yu & Zhihong, 2002) were established using a non-linear hyperplane as a switching manifold. TSMC and FTSMC have proven effective for controlling nonlinear systems under uncertainties and external disturbances. However, their implementation is challenged by a singularity issue near origin caused by negative fractional power terms. In Babes et al. (2021), Corradini and Cristofaro (2018), Li et al. (2022), Rabiee et al. (2019), Su et al. (2023), Tran and Kang (2015), Zuo (2015), researchers proposed new approaches to address this type of singularity, referring to them as non-singular-terminal sliding mode control (NSTSMC). Motivated by the advantages of NSTSMC, the present paper focuses on designing some non-singular terminal sliding surfaces (NSTSS) and corresponding controller.

In recent days, the control of chaotic and hyper-chaotic systems has become a fascinating topic for researchers, as it plays a crucial role in various engineering applications such as secure communication (Pang et al., 2016), biological systems (Blasius & Stone, 2000), robotic systems (Moysis et al., 2020), and cryptography (Nestor et al., 2022). This interest is mainly because of the fact that one of the most important characteristics of chaotic and hyper-chaotic systems is extremely sensitive to the system's initial condition. In 1990, Ott et al. (1990) proposed a control strategy known as the OGY approach which was the first approach to control chaotic systems. Following this, numerous effective ways to controlling chaotic systems are presented in Benkouider et al. (2020), Luo and Zeng (2015), Yau et al. (2000). Moreover, the unpredictable behavior of chaotic systems makes them particularly suitable for applications in secure communication. To implement a chaos-based SCS, two chaotic/hyper-chaotic systems are required: a master system (transmitter) and a slave system (receiver). Some properties of chaotic systems, such as deterministic behavior, can be used for encryption and decryption of messages. The most common techniques utilized are chaotic masking and chaotic modulation. In the present article, the encryption–decryption process is considered as suggested in Benkouider et al. (2020), which is more complex and offers a higher level of security.

It is worth mentioning that in all the existing formulations reported in Hu et al. (2020, 2017), Kong and Zhu (2021), Li et al. (2019, 2020), Polyakov (2011), Xu et al. (2023), the right hand side of Lyapunov inequalities always contains two separate terms, namely $-b_0(\mathcal{V}(X))^\alpha$, $b_0 > 0$, $0 < \alpha < 1$ and $-c_0(\mathcal{V}(X))^\beta$; $c_0 > 0$, $\beta > 1$. When $\mathcal{V}(X) \geq 1$, the second term dominates the dynamics and drives $\mathcal{V}(X)$ toward unity, whereas for $\mathcal{V}(X) < 1$, the first term becomes dominant and ensures convergence to the origin. However, both terms are enforced simultaneously over the entire domain, even in regions where one of them is not required, which leads to unnecessary control action, increased energy consumption, and higher tuning complexity. Moreover, most of the existing articles on

FX_T -S focuses on reducing the upper bound of the settling-time, without concentrating much on the accurate estimation of the settling-time. As a result, a significant gap may arise between the true convergence time and its estimated upper bound, which can restrict practical applicability of the proposed approaches.

Based on the above analysis and to address these issues, this paper proposes a Lyapunov inequality incorporating a Heaviside function to unify the convergence behavior through a single term. Thus, the main contributions of this paper are summarized as follows:

1. Some Lyapunov based sufficient conditions are proposed to achieve FX_T -S of non-linear systems.
2. The notion of Heaviside function is used to reduce the involvement of number of parameters and thus, to reduce the energy consumption.
3. The upper bound of the settling-time estimate corresponding to the proposed results are less conservative in comparison to some of the existing results of similar kind. However, it has been verified numerically that the proposed approach estimates the upper bound to be more accurate as the difference between the true convergence time and the upper bound estimate is shown to be smaller in the present case in comparison to many of the existing articles (Chen et al., 2020; Hu et al., 2020, 2017; Li et al., 2019, 2020; Parsegov et al., 2012; Polyakov, 2011).
4. Based on the aforementioned sufficient conditions, some NSTSS and corresponding controllers are provided. Further, a SCS, based on fixed-time synchronization of hyper-chaotic systems, is described and numerically verified to validate the theoretical results.

The rest of the paper is structured as follows: In Section 2 some notations and preliminaries are given. In Section 3 we present the main result and in Section 4 the application of proposed result. Section 5 contains various numerical simulations to validate the theoretical results. At last in Section 6, conclusion of this paper is given.

2. Notation and preliminaries

Before going to the main section, we will first recollect some definitions and Lemmas that will be used as preliminaries for our results.

2.1. Notation

- $\mathbb{R}$ and $\mathbb{R}^n$ represents a collection of real numbers and an n -dimensional space with the Euclidean norm, respectively. $\mathbb{R}_{\geq 0}$ is a collection of non-negative real numbers.
- $H(v - a_0)$, $v \in \mathbb{R}$ denotes the Heaviside function and defined as $H(v - a_0) = \begin{cases} 1 & ; v \geq a_0 \\ 0 & ; v < a_0 \end{cases}$, $\forall a_0 \in \mathbb{R}$.
- The Hyper-geometric function has been defined as, $F(a_0, b_0; c_0; z) = \frac{\Gamma(c_0)}{\Gamma(b_0)\Gamma(c_0-b_0)} \int_0^1 t^{b_0-1} (1-t)^{c_0-b_0-1} (1-zt)^{-a_0} dt$, where $0 < b_0 < c_0$, $z < 1$.

2.2. Preliminaries

Consider the non-linear system described by the differential equation.

$$\dot{X}(t) = h(X(t)), X(0) = X_0, \quad (1)$$

where $X(t) = (X_1(t), X_2(t), X_3(t), \dots, X_n(t)) \in \mathbb{R}^n$ denotes the state variable of system (1) and $h: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a non-linear function such that $h(0) = 0$, that is, origin $X(t) = 0$ is an equilibrium point of (1). In addition, it is assumed that h meets all of the conditions that guarantee the existence of a unique solution in forward time.

Definition 1. (finite time stability(F_T -S) Bhat & Bernstein, 2000). If for any solution $X(t, X_0)$ of the system (1), the following conditions hold for the origin $X(t) = 0$ of system (1):

1. Lyapunov stability, that is, for every $\chi > 0$, $\exists A(\chi) > 0$ such that $\|X(t)\| < \chi$, $\forall t \geq 0$ whenever $\|X_0\| < A(\chi)$.
2. $\exists$ a function $T : \mathbb{R}^n \setminus \{0\} \rightarrow (0, +\infty)$ such that $\lim_{t \rightarrow T(X_0)} X(t, X_0) = 0$ and $X(t, X_0) = 0 \forall t \geq T(X_0)$. Where T is called the settling time function.

Then the origin $X(t) = 0$ of system (1) is called $F_T S$.

Lemma 1. (Bhat & Bernstein, 2000): Consider the system (1). Assume that there exists a continuous, positive definite and radially unbounded function $\mathcal{V} : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ such that any solution $X(t)$ of system (1) satisfies the inequality

$$\dot{\mathcal{V}}(X) \leq -b_0(\mathcal{V}(X))^\mu, X \in \mathbb{R}^n \setminus \{0\},$$

where $b_0 > 0$, $0 < \mu < 1$. Then, the origin of system (1) is $F_T S$ and the settling time function satisfies

$$T(X_0) \leq -\frac{(\mathcal{V}(X_0))^{(1-\mu)}}{b_0(1-\mu)}.$$

Definition 2. (fixed time stability ($F X_T S$) Polyakov, 2011). The origin $X(t) = 0$ of system (1) is said to be $F X_T S$, if the following conditions hold:

1. Origin $X(t) = 0$ is finite time stable.
2. $\exists T_{\max} > 0$ such that $\forall X_0 \in \mathbb{R}^n \setminus \{0\}$, the settling time function $T(X_0)$ satisfies $T(X_0) \leq T_{\max}$.

Lemma 2. (Li et al., 2019): For system (1), if there exists a regular, positive definite and radially unbounded function $\mathcal{V} : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ such that any solution $X(t)$ of system (1) satisfies the inequality

$$\dot{\mathcal{V}}(X) \leq a_0 \mathcal{V}(X) - b_0(\mathcal{V}(X))^\mu - c_0(\mathcal{V}(X))^\nu,$$

where $a_0, b_0, c_0 > 0$, $0 < \mu < 1 < \nu$ and $a_0 < \min\{b_0, c_0\}$, then, the origin of system (1) is $F X_T S$ and the settling time function satisfies

$$T(X_0) \leq T_{\max}^1 = \frac{1}{(b_0 - a_0)(1 - \mu)} + \frac{1}{(c_0 - a_0)(\nu - 1)}.$$

Lemma 3. (Li et al., 2020): For system (1), if there exists a regular, positive definite and radially unbounded function $\mathcal{V} : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ such that any solution $X(t)$ of system (1) satisfies the inequality

$$\dot{\mathcal{V}}(X) \leq a_0 \mathcal{V}(X) - b_0(\mathcal{V}(X))^\mu - c_0(\mathcal{V}(X))^{\nu + \text{sign}(\mathcal{V}-1)},$$

where $a_0 \in \mathbb{R}$, $b_0, c_0 > 0$, $a_0 < \min\{b_0, c_0\}$, $0 < \mu < 1 < \nu$, then, the origin of system (1) is $F X_T S$ and the settling time function satisfies

$$T(X_0) \leq T_{\max}^2 = \begin{cases} \frac{1}{b_0(1-\mu)} + \frac{1}{vc_0} & ; a_0 = 0 \\ \frac{1}{a_0(1-\mu)} \ln \frac{b_0}{b_0 - a_0} + \frac{1}{vc_0} & ; a_0 < 0. \\ \frac{1}{a_0(1-\mu)} \ln \frac{b_0}{b_0 - a_0} + \frac{1}{v(c_0 - a_0)} & ; a_0 > 0 \end{cases}$$

Lemma 4. (Hardy et al., 1952) Suppose $s_1, s_2, s_3, \dots, s_k \geq 0$, $0 < \mu < 1 < \nu$. Then the following inequalities hold:

$$\sum_{i=1}^k s_i^\mu \geq \left(\sum_{i=1}^k s_i \right)^\mu,$$

$$\sum_{i=1}^k s_i^\nu \geq k^{1-\nu} \left(\sum_{i=1}^k s_i \right)^\nu.$$

3. Main results

Theorem 1. Suppose that a continuous, radially unbounded, positive definite function $\mathcal{V} : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ exists such that the time derivative of $\mathcal{V}$ along the trajectories of (1) conforms to

$$\dot{\mathcal{V}}(X) \leq a_0 \mathcal{V}(X) - b_0(\mathcal{V}(X))^{\mu+\nu \mathcal{H}(\mathcal{V}-1)}, \quad (2)$$

where $b_0 > 0$, $a_0 < b_0$, $0 \leq \mu < 1 < \nu$, then, the origin of system (1) is $F X_T S$ and the estimated upper bound of settling time function $T(x_0)$ is given by

$$T_{\max}^3 = \begin{cases} \frac{1}{b_0(1-\mu)} + \frac{1}{b_0(\mu+\nu-1)} & ; a_0 = 0 \\ \frac{1}{a_0(1-\mu)} \ln \frac{b_0}{b_0 - a_0} + \frac{1}{a_0(\mu+\nu-1)} \ln \frac{b_0}{b_0 - a_0} & ; a_0 \neq 0 \end{cases}. \quad (3)$$

Proof. Assume $\mathcal{V} = \mathcal{V}(X)$ for computational ease. We first establish that the system (1) is $F_T S$ under condition (2).

Case 1: When $0 < \mathcal{V} < 1$, Eq. (2) can be reduced to

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V} - b_0 \mathcal{V}^\mu. \quad (4)$$

If $a_0 \leq 0$, one can obtain from Eq. (4), that

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V} - b_0 \mathcal{V}^\mu \leq -b_0 \mathcal{V}^\mu < 0.$$

If $a_0 > 0$, Eq. (4) reduces to

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V} - b_0 \mathcal{V}^\mu < -(b_0 - a_0) \mathcal{V}^\mu < 0.$$

Case 2: When $\mathcal{V} \geq 1$, one can obtain from equation (2)

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V} - b_0 \mathcal{V}^{\mu+\nu}. \quad (5)$$

If $a_0 \leq 0$, from Eq. (5), it is easy to calculate that

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V} - b_0 \mathcal{V}^{\mu+\nu} \leq -b_0 \mathcal{V}^{\mu+\nu} < -b_0 \mathcal{V}^\mu < 0.$$

And, if $a_0 > 0$, one can obtain from Eq. (5), that

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V} - b_0 \mathcal{V}^{\mu+\nu} < -(b_0 - a_0) \mathcal{V}^{\mu+\nu} < -(b_0 - a_0) \mathcal{V}^\mu < 0.$$

Based on the arguments above and Lemma 1, one can conclude that the origin of system (1) is $F_T S$. Furthermore, from Eq. (2), one can easily obtain that

$$T(X_0) \leq \int_0^{\mathcal{V}_0} \frac{ds}{-a_0 s + b_0 s^{\mu+\nu \mathcal{H}(s-1)}}, \quad (6)$$

where $\mathcal{V}(0) = \mathcal{V}_0$ and $T(X_0)$ is the settling time function of system (1). Now, in order to find an upper bound of the settling time, we consider the following two cases,

Case 1: If $0 < \mathcal{V}_0 < 1$, from Eq. (6), one can obtain that

$$\begin{aligned} T(X_0) &\leq \int_0^{\mathcal{V}_0} \frac{ds}{-a_0 s + b_0 s^{\mu+\nu \mathcal{H}(s-1)}} \\ &\leq \int_0^1 \frac{ds}{-a_0 s + b_0 s^\mu} \\ &= \int_0^1 \frac{s^{-\mu} ds}{-a_0 s^{1-\mu} + b_0}. \end{aligned} \quad (7)$$

Let $z = s^{1-\mu}$, from Eq. (7), one obtains that

$$T(X_0) \leq \frac{1}{(1-\mu)} \int_0^1 \frac{dz}{-a_0 z + b_0}. \quad (8)$$

If $a_0 = 0$, one can easily get from Eq. (8) that

$$T(X_0) \leq \frac{1}{b_0(1-\mu)}. \quad (9)$$

If $a_0 \neq 0$, from Eq. (8)

$$T(X_0) \leq \frac{1}{a_0(1-\mu)} \ln \frac{b_0}{b_0 - a_0}. \quad (10)$$

Case 2: If $\mathcal{V}_0 \geq 1$, one can get from equation (6)

$$\begin{aligned} T(X_0) &\leq \int_0^{\mathcal{V}_0} \frac{ds}{-a_0 s + b_0 s^{\mu+\nu \mathcal{H}(s-1)}} \\ &\leq \int_0^1 \frac{ds}{-a_0 s + b_0 s^\mu} + \int_1^\infty \frac{ds}{-a_0 s + b_0 s^{\mu+\nu}}. \end{aligned} \quad (11)$$

First integral in Eq. (11) is same as in Case 1. For the second integral, letting $z = s^{1-\mu-\nu}$ one obtains

$$\int_1^\infty \frac{ds}{-a_0 s + b_0 s^{\mu+\nu}} = \frac{1}{(\mu + \nu - 1)} \int_0^1 \frac{dz}{-a_0 z + b_0}. \quad (12)$$

If $a_0 = 0$, one can easily get from Eq. (12)

$$\int_1^\infty \frac{ds}{-a_0 s + b_0 s^{\mu+\nu}} = \frac{1}{b_0(\mu + \nu - 1)}. \quad (13)$$

If $a_0 \neq 0$, from Eq. (12)

$$\int_1^\infty \frac{ds}{-a_0 s + b_0 s^{\mu+\nu}} = \frac{1}{a_0(\mu + \nu - 1)} \ln \frac{b_0}{b_0 - a_0}. \quad (14)$$

From Eqs. (9),(10),(13) and (14), one can deduce that the settling time satisfies

$$T(X_0) \leq T_{\max}^3,$$

where

$$T_{\max}^3 = \begin{cases} \frac{1}{b_0(1-\mu)} + \frac{1}{b_0(\mu+\nu-1)} & ; a_0 = 0 \\ \frac{1}{a_0(1-\mu)} \ln \frac{b_0}{b_0-a_0} + \frac{1}{a_0(\mu+\nu-1)} \ln \frac{b_0}{b_0-a_0} & ; a_0 \neq 0 \end{cases}.$$

The mathematical proof of Theorem 1 has been completed. $\square$

Remark 1. Unlike the forms given in articles (Chen et al., 2020; Hu et al., 2020, 2017; Li et al., 2019, 2020; Parsegov et al., 2012; Polyakov, 2011), Theorem 1, as established above does not contain two terms $-b_0(\mathcal{V}(X))^{\mu}$, $b_0 > 0, 0 < \mu < 1$ and $-c_0(\mathcal{V}(X))^{\nu}$, $c_0 > 0, \nu > 1$ individually. Instead, only one term $-b_0(\mathcal{V}(X))^{\mu+\nu.H(\mathcal{V}-1)}$, $b_0 > 0, 0 \leq \mu < 1 < \nu$ which represents the relation between $\mathcal{V}$ and 1. Moreover, this term plays the role of both terms as follows: when $\mathcal{V} \geq 1$ the term $-b_0(\mathcal{V}(X))^{\mu+\nu.H(\mathcal{V}-1)}$ reduces to $-b_0(\mathcal{V}(X))^{\mu+\nu}$, $\mu + \nu > 1$, ensuring that $\mathcal{V}(t)$ decreases toward 1. Subsequently, when $\mathcal{V}(t) < 1$, the same term $-b_0(\mathcal{V}(X))^{\mu+\nu.H(\mathcal{V}-1)}$ reduces to $-b_0(\mathcal{V}(X))^{\mu}$, $0 \leq \mu < 1$, thereby guaranteeing that $\mathcal{V}(t)$ further decreases to zero. With this Heaviside-based formulation, the convergence analysis avoids the simultaneous enforcement of multiple terms over the entire domain, thereby reducing conservatism and leading to a smaller gap between the true convergence time and its upper-bound estimate.

Corollary 1. Consider the non-linear scalar system

$$\dot{X} = -(a_0|X| + b_0|X|^{\mu+\nu.H(|X|-1)}).sign(X) \quad (15)$$

where $a_0, b_0 > 0, 0 \leq \mu < 1 < \nu$. Then, the origin of system (15) is FX_{T-S} and the estimated upper bound of settling time function $T(X_0)$ is given by

$$T_{\max} = \frac{1}{a_0(1-\mu)} \ln(1 + \frac{a_0}{b_0}) + \frac{1}{a_0(\mu + \nu - 1)} \ln(1 + \frac{a_0}{b_0}).$$

Remark 2. The proposed approach achieves better accuracy than some previous approaches reported in Chen et al. (2020), Hu et al. (2017), Parsegov et al. (2012), Polyakov (2011). Specifically, estimation of the settling time is smaller than that as obtained in the aforementioned literature. To ensure a fair comparison, we will be considering the scalar differential equations and corresponding estimated upper bound as shown in Table 1.

Fig. 1 shows the trajectories of $X(t)$, obtained by solving the differential equations listed in Table 1 by considering identical initial condition and parameter values as follows: $X(0) = 3, a_0 = 1, b_0 = 2, c_0 = 2, \mu = \frac{3}{4}$ and $\nu = \frac{5}{4}$. From Fig. 1, it is confirmed that the estimated upper bound in this paper is smaller than in Chen et al. (2020), Hu et al. (2017), Parsegov et al. (2012), Polyakov (2011). In addition, Fig. 1 clearly shows that the gap between the true convergence time and its upper bound (magenta curve) is smaller in the present work as compared to the others, which imply a better accuracy of the present result.

Remark 3. If $b_0 = c_0$ and $a_0 \geq 0$, then $T_{\max}^3 < T_{\max}^1$.

Proof. When $a_0 = 0$, from Eq. (3)

$$\begin{aligned} T_{\max}^3 &= \frac{1}{b_0(1-\mu)} + \frac{1}{b_0(\mu+\nu-1)} \\ &< \frac{1}{b_0(1-\mu)} + \frac{1}{b_0(\nu-1)} \\ &= \frac{1}{b_0(1-\mu)} + \frac{1}{c_0(\nu-1)} = T_{\max}^1. \end{aligned} \quad (16)$$

When $a_0 > 0$, again from Eq. (3)

$$T_{\max}^3 = \frac{1}{a_0(1-\mu)} \ln \frac{b_0}{b_0-a_0} + \frac{1}{a_0(\mu+\nu-1)} \ln \frac{b_0}{b_0-a_0}. \quad (17)$$

We know that, $\ln(1+x) < x, \forall x > 0$, let $x = \frac{a_0}{b_0-a_0}$, from Eq. (17) we obtain

$$\begin{aligned} T_{\max}^3 &< \frac{1}{(b_0-a_0)(1-\mu)} + \frac{1}{(b_0-a_0)(\mu+\nu-1)} \\ &< \frac{1}{(b_0-a_0)(1-\mu)} + \frac{1}{(b_0-a_0)(\nu-1)} \\ &= \frac{1}{(b_0-a_0)(1-\mu)} + \frac{1}{(c_0-a_0)(\nu-1)} = T_{\max}^1. \end{aligned} \quad (18)$$

It follows from Eqs. (16) and (18) that $T_{\max}^3 < T_{\max}^1$. That is, estimated upper bound of the settling time of Theorem 1 is smaller than that of Lemma 2. $\square$

Remark 4. Fig. 2 shows the trajectories of $X(t)$, obtained by solving the differential equations listed in Table 2 by taking identical initial condition and parameters values. From Fig. 2 and Table 2, it is confirmed numerically that the difference between the true convergence time and its upper bound in Theorem 1 is smaller than that in Lemmas 2 and 3. Thus the proposed approach achieves better accuracy than that in Li et al. (2019), Li et al. (2020).

Remark 5. To analyze the impact of parameters a_0, b_0, μ and ν on $T_{\max}^3$, calculate the first derivative of $T_{\max}^3$ with respect to a_0, b_0, μ and ν separately, to obtain

$$\frac{dT_{\max}^3}{da_0} = \frac{\nu}{a_0^2(1-\mu)(\mu+\nu-1)} \left(\frac{a_0}{b_0-a_0} - \ln \frac{a_0}{b_0-a_0} \right) > 0,$$

$$\frac{dT_{\max}^3}{db_0} = \frac{-\nu}{b_0(b_0-a_0)(1-\mu)(\mu+\nu-1)} < 0,$$

$$\frac{dT_{\max}^3}{d\mu} = \frac{\nu(\nu-2+2\mu)}{a_0(1-\mu)^2(\mu+\nu-1)^2} \ln \frac{b_0}{b_0-a_0}$$

$$\frac{dT_{\max}^3}{d\nu} = \frac{-1}{a_0(\mu+\nu-1)^2} \ln \frac{b_0}{b_0-a_0} < 0.$$

From the above derivatives, it is clear that $T_{\max}^3$ will be enhanced with larger values of a_0 , smaller values of b_0 and ν , while there is no clear relationship between $T_{\max}^3$ and μ .

Remark 6. If we replace $H(\mathcal{V}-1)$ with more general term $H(\mathcal{V}-k)$, $k > 0$ and other parameters remain the same as in Theorem 1, then we have

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V} - b_0 \mathcal{V}^{\mu+\nu.H(\mathcal{V}-k)}. \quad (19)$$

and

$$T(X_0) \leq T_{\max}^3 = \begin{cases} \frac{k^{(1-\mu)}}{b_0(1-\mu)} + \frac{1}{b_0 k^{(\mu+\nu-1)(\mu+\nu-1)}} & ; a_0 = 0 \\ \frac{k^{(1-\mu)}}{a_0(1-\mu)} \ln \frac{b_0}{b_0-a_0} + \frac{1}{a_0 k^{(\mu+\nu-1)(\mu+\nu-1)}} \ln \frac{b_0}{b_0-a_0} & ; a_0 \neq 0 \end{cases}. \quad (20)$$

The proof is similar to that in Theorem 1.

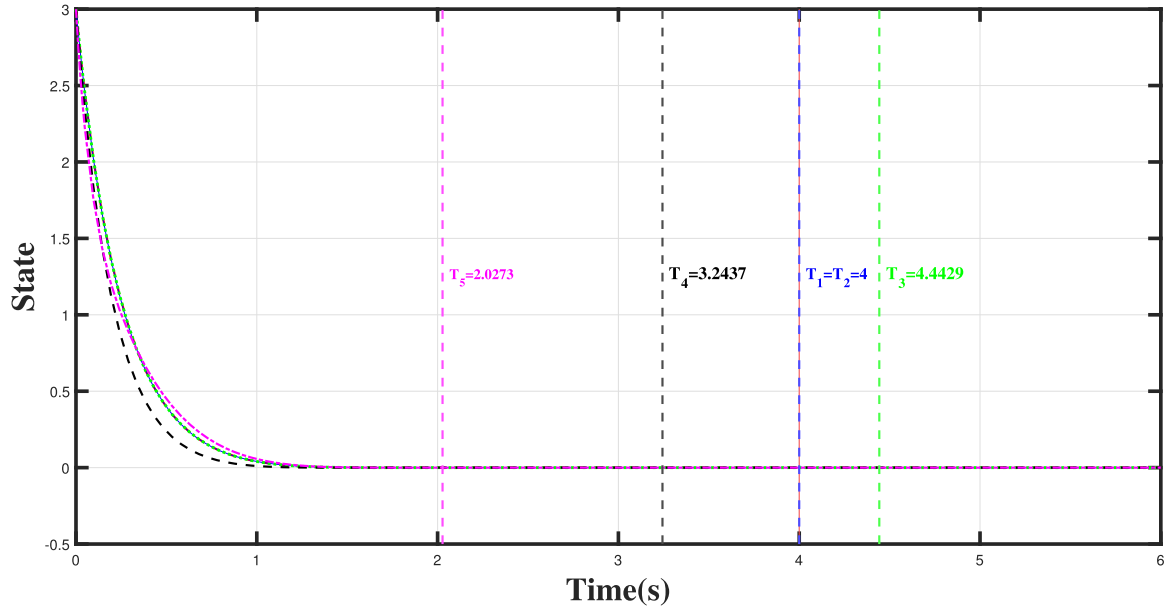
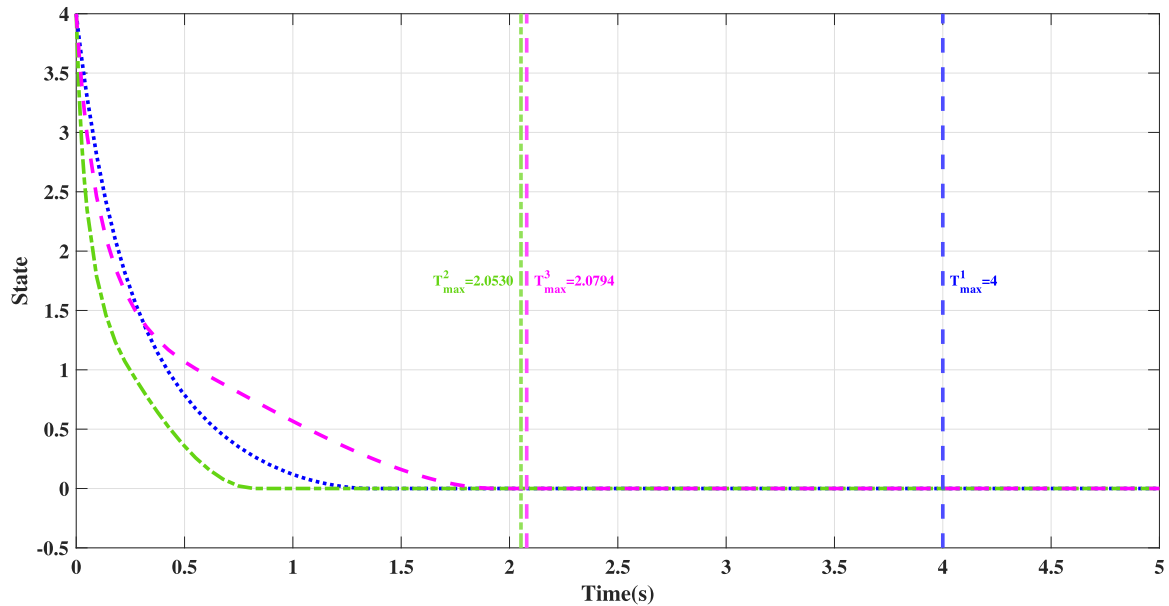
Next, we find the optimal value of k for which $T_{\max}^3$ is minimum as follows:

$$\frac{dT_{\max}^3}{dk} = \begin{cases} \frac{k^{-\mu}}{b_0} + \frac{k^{-(\mu+\nu)}}{b_0(\mu+\nu-1)} & ; a_0 = 0 \\ \frac{k^{-\mu}}{a_0} \ln \frac{b_0}{b_0-a_0} + \frac{k^{-(\mu+\nu)}}{a_0} \ln \frac{b_0}{b_0-a_0} & ; a_0 \neq 0 \end{cases}. \quad (21)$$

Table 1

Comparison of existing results and the proposed method.

Ref.	$\dot{X} = h(X)$	Upper bound T
Polyakov, 2011	$\dot{X} = -(b_0 X ^\mu + c_0 X ^\nu).sign(X); b_0, c_0 > 0, 0 < \mu < 1 < \nu$	$T_1 = \frac{1}{b_0(1-\mu)} + \frac{1}{c_0(\nu-1)}$
Hu et al., 2017	$\dot{X} = -(b_0 X ^\mu + c_0 X ^\nu).sign(X); b_0, c_0 > 0, 0 < \mu < 1 < \nu$	$T_2 = \frac{1}{c_0} \left(\frac{c_0}{b_0} \right)^{\frac{1-\mu}{\nu-\mu}} \left[\frac{1}{(1-\mu)} + \frac{1}{(\nu-1)} \right]$
Parsegov et al., 2012	$\dot{X} = -(b_0 X ^\mu + c_0 X ^\nu).sign(X); b_0, c_0 > 0, 0 \leq \mu < 1 < \nu$	$T_3 = \frac{x m_0}{\sqrt{b_0 c_0}}; \mu = 1 - \frac{1}{2m_0}, \nu = 1 + \frac{1}{2m_0}, m_0 > 1$
Chen et al., 2020	$\dot{X} = -(a_0 X + b_0 X ^\mu + c_0 X ^\nu).sign(X); a_0, b_0, c_0 > 0, 0 \leq \mu < 1 < \nu$	$T_4 = \frac{1}{a_0(1-\mu)} \ln(1 + \frac{a_0}{b_0}) + \frac{1}{a_0(\nu-1)} \ln(1 + \frac{a_0}{c_0})$
This paper (Corollary 1)	$\dot{X} = -(a_0 X + b_0 X ^{\mu+\nu H(X ^{-1})}).sign(X); a_0, b_0 > 0, 0 \leq \mu < 1 < \nu$	$T_5 = \frac{1}{a_0(1-\mu)} \ln(1 + \frac{a_0}{b_0}) + \frac{1}{a_0(\mu+\nu-1)} \ln(1 + \frac{a_0}{b_0})$

**Fig. 1.** Time evolution of $X(t)$ based on the differential equations listed in Table 1 and corresponding estimated upper bound of settling time.**Fig. 2.** Time evolution of $X(t)$ based on the differential equations listed in Table 2 and corresponding estimated upper bound $T_{\max}$.**Table 2**Comparative analysis of true convergence time, estimated upper bounds, and the corresponding gap for existing results and the proposed method under identical conditions. Parameters used in the comparison: $a_0 = 1, b_0 = 2, c_0 = 2, \mu = 0.5, \nu = 1.5$ and initial condition $X(0) = 4$.

Ref.	$\dot{X} = h(X)$	Estimated Upper Bounds $T_{\max}$	True Convergence Time t	Gap $ T_{\max} - t $
Li et al., 2019	$\dot{X} = (a_0 X - b_0 X ^\mu - c_0 X ^\nu).sign(X)$	$T_{\max}^1 = 4$	1.34030	2.6597
Li et al., 2020	$\dot{X} = (a_0 X - b_0 X ^\mu - c_0 X ^{\nu+sign(X ^{-1})}).sign(X)$	$T_{\max}^2 = 2.0530$	0.82067	1.2323
Proposed method (Theorem 1)	$\dot{X} = (a_0 X - b_0 X ^{\mu+\nu H(X ^{-1})}).sign(X)$	$T_{\max}^3 = 2.0794$	1.92429	0.1552

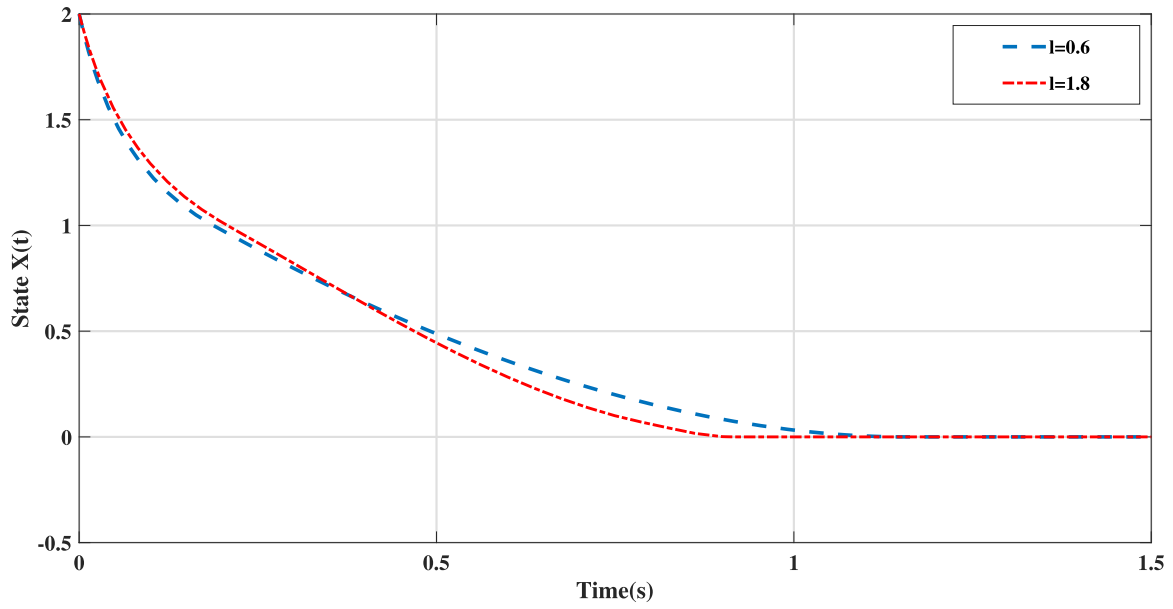


Fig. 3. The state trajectories of system (37) for two values $l = 0.6$, and $l = 1.8$ with initial condition $X(0) = 2$.

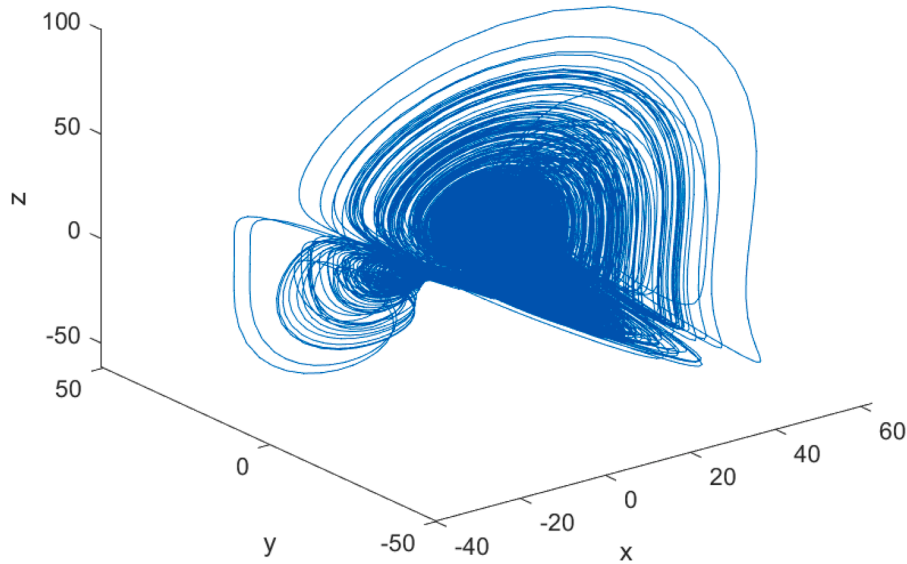


Fig. 4. Trajectories of hyper-chaotic system (38) with initial condition (1,2,0,1).

When $\frac{dT_{\max}^3}{dk} = 0$, in both cases $a_0 = 0$ and $a_0 \neq 0$ we obtain $k = 1$. Further $\frac{d^2T_{\max}^3}{dk^2} > 0$ at $k = 1$. Thus at $k = 1$, $T_{\max}^3$ is minimum.

Theorem 2. Suppose that a continuous, radially unbounded, positive definite function $\mathcal{V} : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ exists such that the time derivative of $\mathcal{V}$ along the trajectories of (1) conforms to

$$\dot{\mathcal{V}} \leq \begin{cases} a_0 \mathcal{V} - b_0 \mathcal{V}^{\mu+\nu} H(\mathcal{V}-1) & ; 0 < \mathcal{V} < 1 \\ a_0 \mathcal{V} - c_0 \mathcal{V}^{\mu+\nu} H(\mathcal{V}-1) & ; \mathcal{V} \geq 1 \end{cases}, \quad (22)$$

where $b_0, c_0 > 0, a_0 < \min(b_0, c_0)$, $0 \leq \mu < 1 < \nu$, then, the origin of system (1) is fixed time stable and the estimated upper bound of settling time is given by

$$T_{\max}^4 = \begin{cases} \frac{1}{b_0(1-\mu)} + \frac{1}{c_0(\mu+\nu-1)} & ; a_0 = 0 \\ \frac{1}{a_0(1-\mu)} \ln \frac{b_0}{b_0-a_0} + \frac{1}{a_0(\mu+\nu-1)} \ln \frac{c_0}{c_0-a_0} & ; a_0 \neq 0 \end{cases}. \quad (23)$$

Proof. The proof does not require any extra complexity and is similar to that of Theorem 1. $\square$

Theorem 3. Suppose that a continuous, radially unbounded, positive definite function $\mathcal{V} : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ exists such that the time derivative of $\mathcal{V}$ along the trajectories of (1) conforms to

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V}^l - b_0 \mathcal{V}^{\mu+\nu} H(\mathcal{V}-1), \quad (24)$$

where $b_0 > 0, a_0 < b_0$, $0 \leq \mu < 1 < \nu$ and $\mu \leq l \leq \mu + \nu$, then, the origin of system (1) is $FX_T S$ and the estimated upper bound of settling time is given by

$$T(X_0) \leq T_{\max}^5 = \begin{cases} T_1' + T_2 & ; l = \mu \\ T_1 + T_2 & ; \mu < l < \mu + \nu, \\ T_1 + T_2' & ; l = \mu + \nu \end{cases} \quad (25)$$

where,

$$T_1 = \frac{1}{b_0(1-\mu)} \cdot F\left(1, \frac{1-\mu}{l-\mu}; \frac{1+l-2\mu}{l-\mu}; \frac{a_0}{b_0}\right),$$

$$T_2 = \frac{1}{b_0(\mu+\nu-1)} \cdot F\left(1, \frac{\mu+\nu-1}{\mu+\nu-l}; \frac{2(\mu+\nu)-(l+1)}{\mu+\nu-l}; \frac{a_0}{b_0}\right)$$

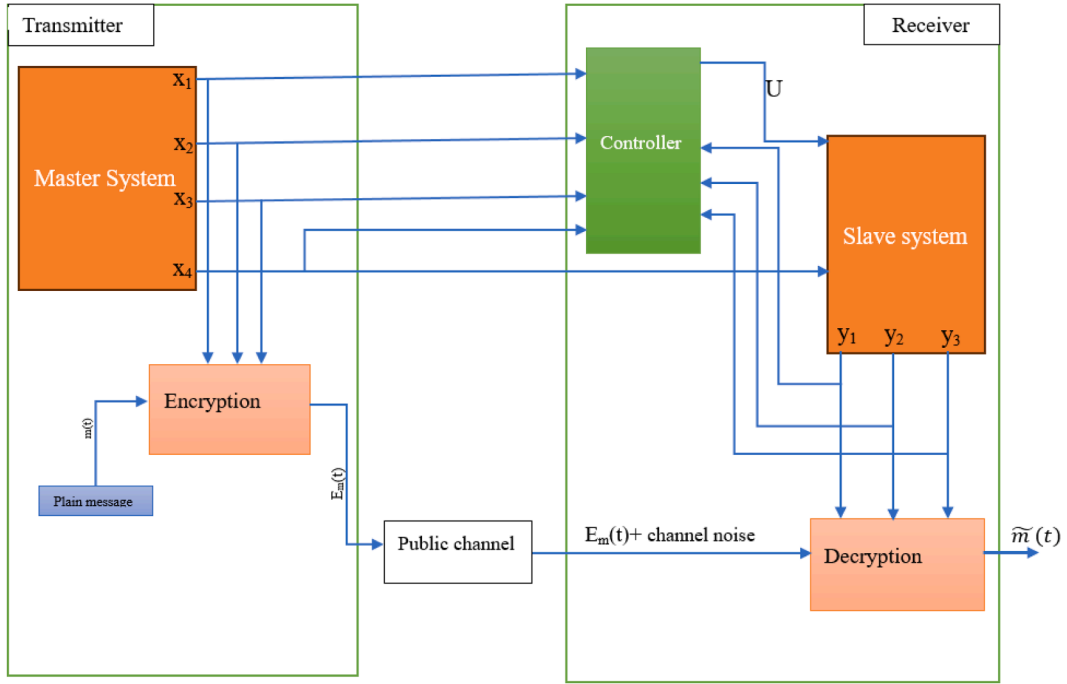


Fig. 5. Secure communication scheme.

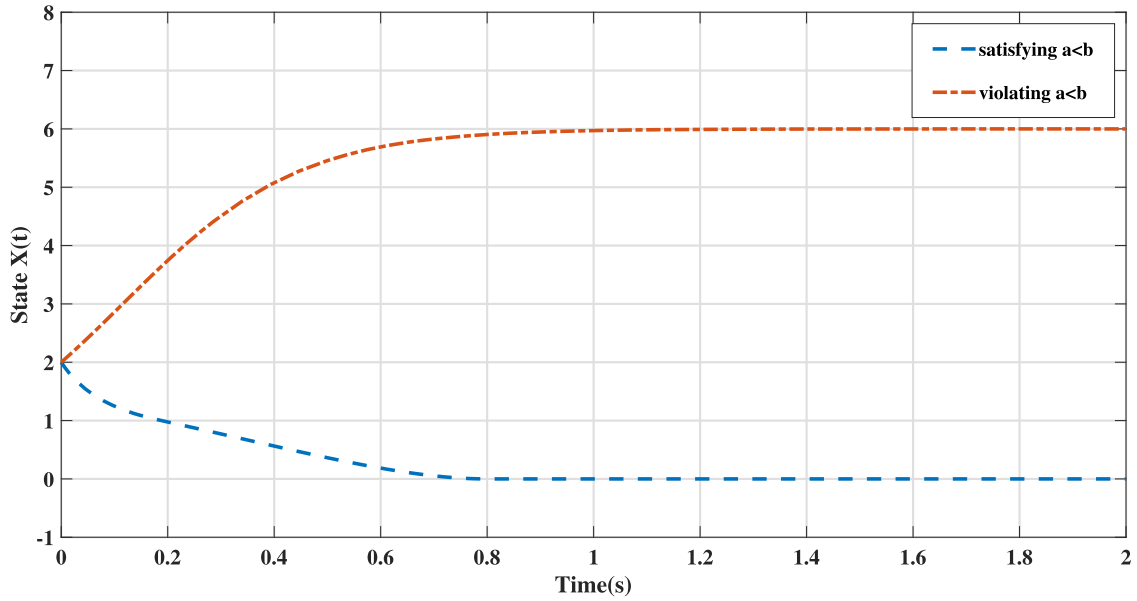


Fig. 6. State trajectories of Eq. (61) with two different cases.

$$T'_1 = \frac{1}{(b_0 - a_0)(1 - l)},$$

$$T'_2 = \frac{1}{(b_0 - a_0)(l - 1)}.$$

Proof. Firstly, we show the $F_T S$ of system(1) under the condition (24).

Case 1: When $0 < \mathcal{V} < 1$, Eq. (24) can be reduced to

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V}^l - b_0 \mathcal{V}^\mu. \quad (26)$$

If $a_0 \leq 0$, one can obtain from Eq. (26), that

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V}^l - b_0 \mathcal{V}^\mu \leq -b_0 \mathcal{V}^\mu < 0.$$

If $a_0 > 0$, Eq. (26) reduces to

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V}^l - b_0 \mathcal{V}^\mu < -(b_0 - a_0) \mathcal{V}^\mu < 0.$$

Case 2: When $\mathcal{V} \geq 1$, one can obtain from equation (24)

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V}^l - b_0 \mathcal{V}^{\mu+\nu}. \quad (27)$$

If $a_0 \leq 0$, from Eq. (27), it is easy to calculate that

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V}^l - b_0 \mathcal{V}^{\mu+\nu} \leq -b_0 \mathcal{V}^{\mu+\nu} < -b_0 \mathcal{V}^\mu < 0.$$

And, if $a_0 > 0$, one can obtain from Eq. (27), that

$$\dot{\mathcal{V}} \leq a_0 \mathcal{V}^l - b_0 \mathcal{V}^{\mu+\nu} < -(b_0 - a_0) \mathcal{V}^{\mu+\nu} < -(b_0 - a_0) \mathcal{V}^\mu < 0.$$

Based on the arguments above and Lemma 1, one can conclude that the origin of system (1) is $F_T S$.

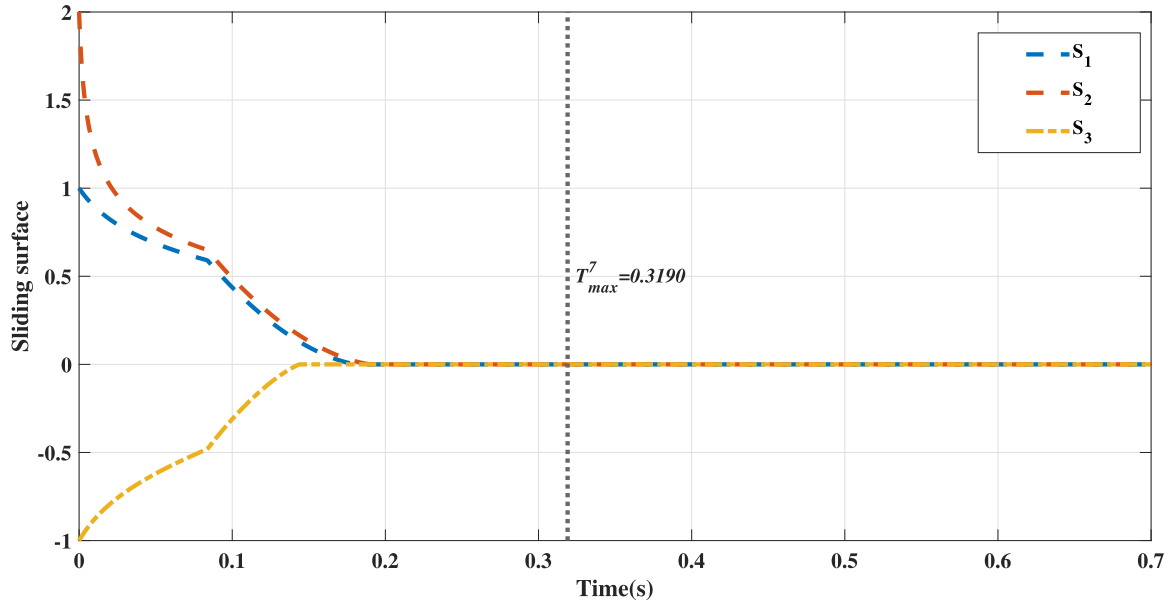
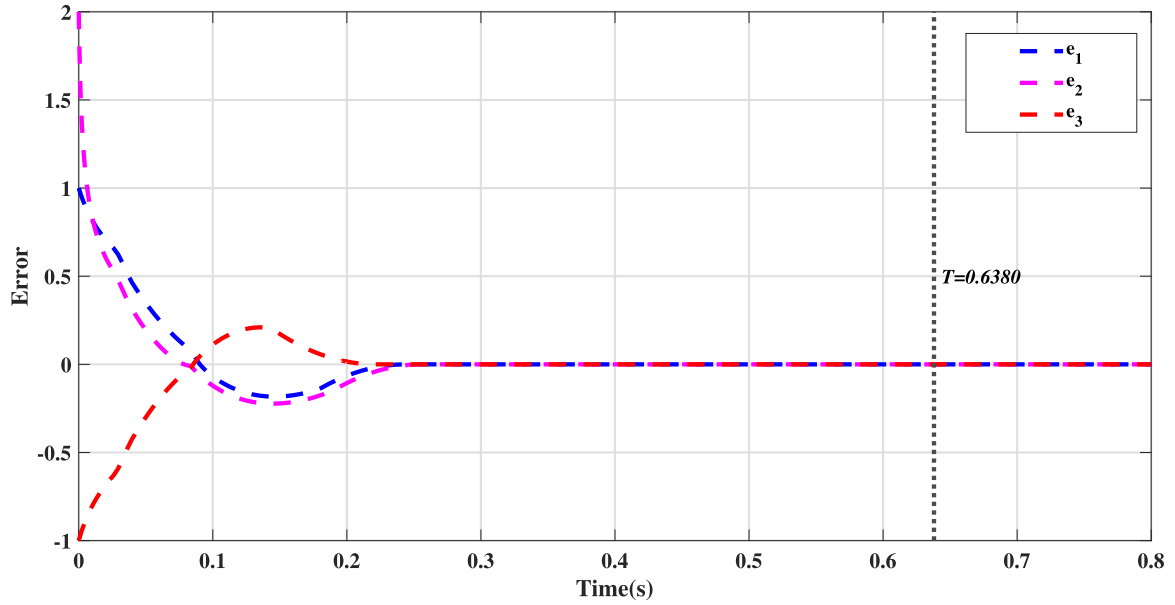


Fig. 7. Stability of Sliding surfaces of Eq. (44).

Fig. 8. The state trajectories of the error system (42) when $a_1 = a_2 = -1$.

Furthermore, from Eq. (24), one can easily obtain that

$$T(X_0) \leq \int_0^{\mathcal{V}_0} \frac{ds}{-a_0 s^l + b_0 s^{\mu+\nu} H(s-1)}.$$

Case 1: If $0 < \mathcal{V}_0 < 1$, from Eq. (28), one can obtain that

$$T(X_0) \leq \int_0^1 \frac{ds}{-a_0 s^l + b_0 s^{\mu}}.$$

If $\mu < l \leq \mu + \nu$, from Eq. (29), letting

$$\begin{aligned} T_1 &= \int_0^1 \frac{ds}{-a_0 s^l + b_0 s^{\mu}} \\ &= \frac{1}{b_0} \int_0^1 \frac{s^{-\mu} ds}{1 - \frac{a_0}{b_0} s^{l-\mu}}. \end{aligned}$$

Let $z = s^{l-\mu}$, one can easily get from Eq. (30) that

$$(28) \quad T_1 = \frac{1}{b_0(l-\mu)} \int_0^1 \frac{z^{\frac{1-l}{l-\mu}} dz}{1 - \frac{a_0}{b_0} z}$$

$$\begin{aligned} (29) \quad &= \frac{1}{b_0(l-\mu)} \int_0^1 z^{\frac{1-l}{l-\mu}-1} \cdot (1 - \frac{a_0}{b_0} z)^{-1} dz \\ &= \frac{1}{b_0(1-\mu)} \cdot F(1, \frac{1-\mu}{l-\mu}; \frac{1+l-2\mu}{l-\mu}; \frac{a_0}{b_0}). \end{aligned} \quad (31)$$

If $\mu = l$, from Eq. (29),

$$(30) \quad T_1' = \int_0^1 \frac{1}{(b_0 - a_0)s^l} = \frac{1}{(b_0 - a_0)(1-l)}. \quad (32)$$

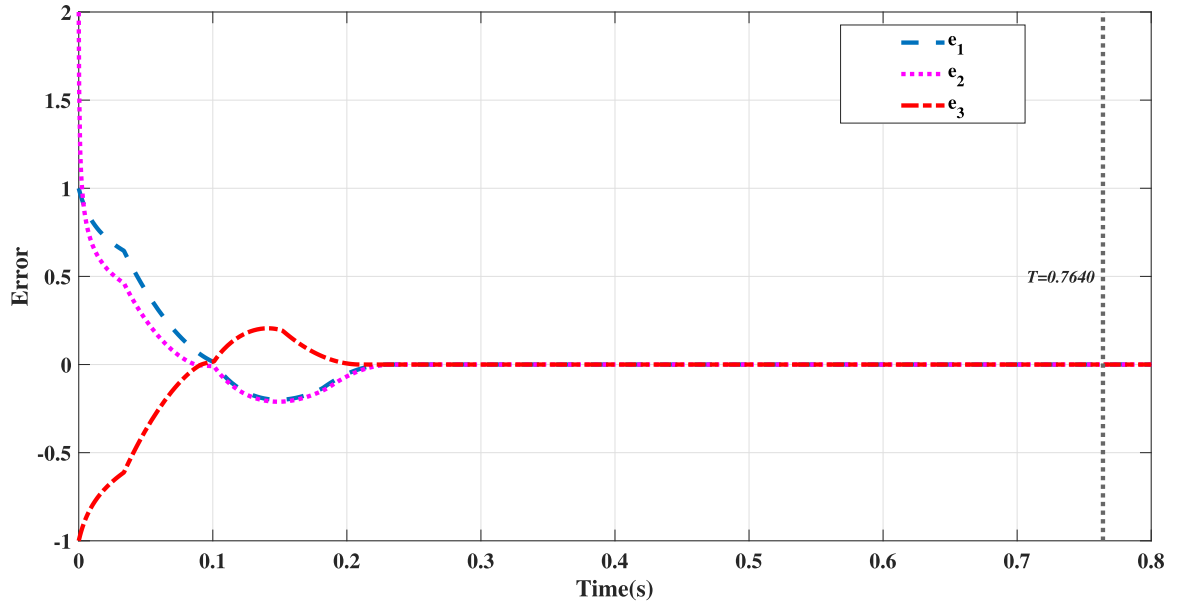


Fig. 9. The state trajectories of the error system (42) when $a_1 = a_2 = 0$.

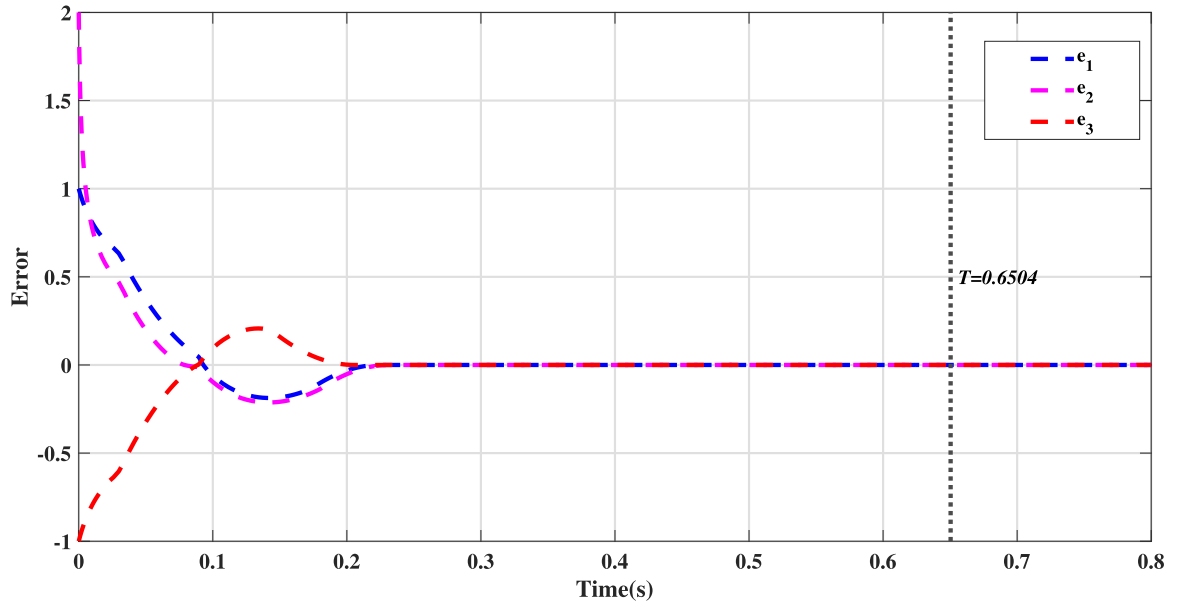


Fig. 10. The state trajectories of the error system (42) when $a_1 = a_2 = 2$.

Case 2: If $\mathcal{V}_0 \geq 1$, from Eq. (28), one can obtain that

$$T(X_0) \leq \int_0^1 \frac{ds}{-a_0 s^l + b_0 s^\mu} + \int_1^\infty \frac{ds}{-a_0 s^l + b_0 s^{\mu+\nu}}. \quad (33)$$

If $\mu \leq l < \mu + \nu$, from Eq. (33), letting

$$\begin{aligned} T_2 &= \int_1^\infty \frac{ds}{-a_0 s^l + b_0 s^{\mu+\nu}} \\ &= \frac{1}{b_0} \int_1^\infty \frac{s^{-(\mu+\nu)} ds}{1 - \frac{a_0}{b_0} s^{l-\mu-\nu}}. \end{aligned} \quad (34)$$

Let $z = s^{l-\mu-\nu}$, one can obtain from Eq. (34) that

$$T_2 = \frac{1}{b_0(l-\mu-\nu)} \int_1^0 \frac{z^{\frac{1-l}{l-\mu-\nu}} dz}{1 - \frac{a_0}{b_0} z}$$

$$\begin{aligned} &= \frac{1}{b_0(\mu+\nu-l)} \int_0^1 z^{\frac{\mu+\nu-l}{\mu+\nu-l}-1} \left(1 - \frac{a_0}{b_0} z\right)^{-1} dz \\ &= \frac{1}{b_0(\mu+\nu-l)} F\left(1, \frac{\mu+\nu-l}{\mu+\nu-l}; \frac{2(\mu+\nu)-(1+l)}{\mu+\nu-l}; \frac{a_0}{b_0}\right). \end{aligned} \quad (35)$$

If $l = \mu + \nu$, from Eq. (33), letting

$$T'_2 = \int_1^\infty \frac{1}{(b_0 - a_0)s^l} = \frac{1}{(b_0 - a_0)(l-1)}. \quad (36)$$

The mathematical proof of Theorem 3 has been completed. $\square$

Remark 7. For $l = 1, T_{\max}^3 = T_{\max}^5$

Remark 8. In most of the articles (Chen et al., 2020; Hu et al., 2020; Kong & Zhu, 2021; Kong et al., 2021; Li et al., 2019, 2020) and Theorem 1, the exponent of $a_0 \mathcal{V}$ is fixed at 1. However, in Theorem 3, we extend the exponent l of $a_0 \mathcal{V}$ to a range between μ and ν . This generaliza-

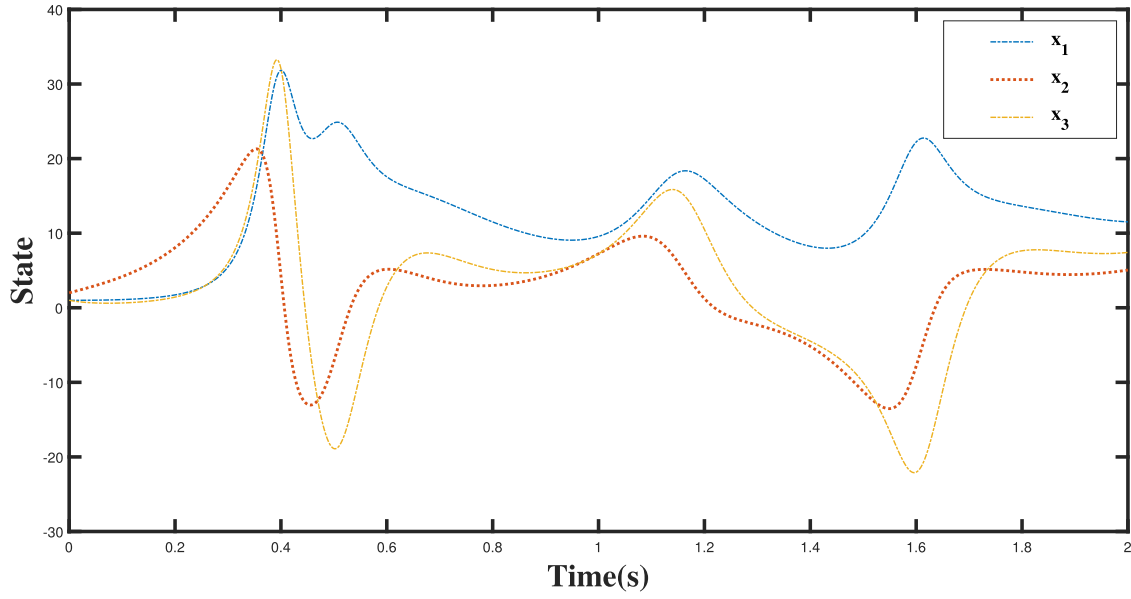


Fig. 11. State trajectories x_1, x_2 and x_3 of Eq. (39).

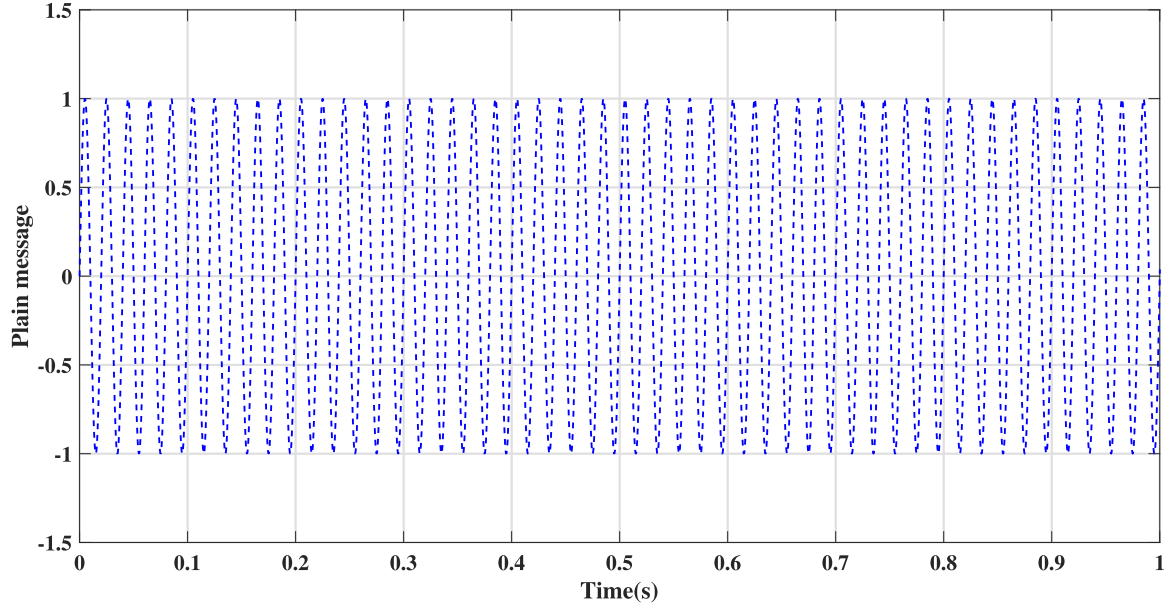


Fig. 12. Evolution of Plain message $m(t) = \sin(100\pi t)$.

tion enhances flexibility and broadens applicability in various practical engineering problems.

Remark 9. For intuitive explanation of the effect of the term $a_0 \mathcal{V}^l$ on the convergence rate, let us consider the scalar system

$$\dot{X} = (a_0 |X|^l - b_0 |X|^{\mu+\nu} H(|X|-1)) \cdot \text{sign}(X) \quad (37)$$

with $a_0 = 1.1, b_0 = 3, \mu = 0.5, \nu = 2$. By considering the Lyapunov function $\mathcal{V}(X) = |X|$, we can know that the system (37) is fixed time stable within $T_{\max}^5$ from Theorem 3. Fig. 3 shows the state trajectories of system (37) for two values $l = 0.6$ and $l = 1.8$.

Next, we provide the theoretical interpretation of Fig. 3. When the initial condition satisfies $|X(0)| > 1$, the state trajectories corresponding to $l = 0.6$ exhibit faster convergence compared to those with $l = 1.8$, since $|X(0)|^{0.6} < |X(0)|^{1.8}$. In contrast for $|X(0)| < 1$, the trajectories with $l = 1.8$ converge faster because $|X(0)|^{0.6} > |X(0)|^{1.8}$. Therefore, the term $a_0 |X|^l$ significantly influences the convergence rate of the system. That

is, this generalization enables better control over the convergence rate of the system.

4. Application in secure communication

4.1. System description: a hyper chaotic system

Consider the hyper-chaotic system given in the paper (Wan et al., 2014)

$$\begin{cases} \dot{x} = -ax + yz + y \\ \dot{y} = cy - xz + dz - w \\ \dot{z} = bxy - hz + yw \\ \dot{w} = y \end{cases} \quad (38)$$

where $X = [x, y, z, w] \in \mathbb{R}^4$ is the system state and $a, b, c, d, h \in \mathbb{R}^+$ are the system parameters. This system generates the hyper-chaotic behav-

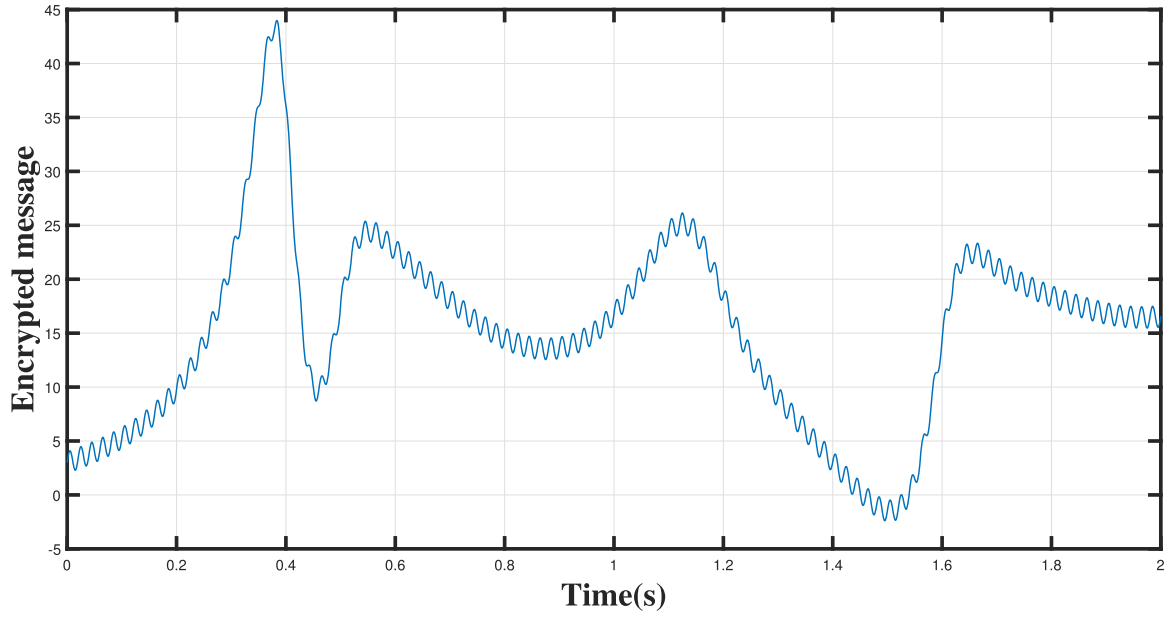
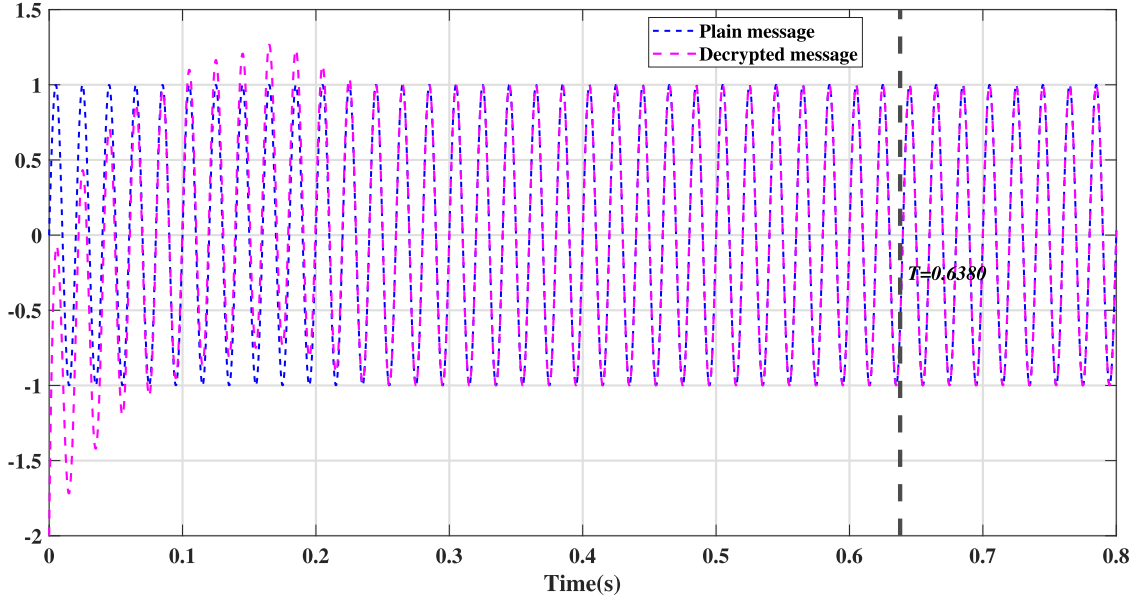


Fig. 13. Evolution of Encrypted message over time.

Fig. 14. Decryption and Retrieval of the plain message at receiver end with in a time $T = 0.6380$.

ior for different values of a, b, c, d, h . Here we consider the values of these parameters as $a = 4, b = 2, c = 5.5, d = 10$ and $h = 19$, then system generates hyper-chaotic attractor as shown in Fig. 4.

Remark 10. Lyapunov exponents of considered hyper-chaotic system are $+2.108, +2.108, 0$, and -19.1293 as mentioned in paper (Wan et al., 2014). It indicates a higher level of complexity and unpredictability in the system's behavior.

4.2. Scheme description

In this section, a brief description of proposed scheme for secure communication based on above hyper-chaotic system (38) is provided and depicted in Fig. 5.

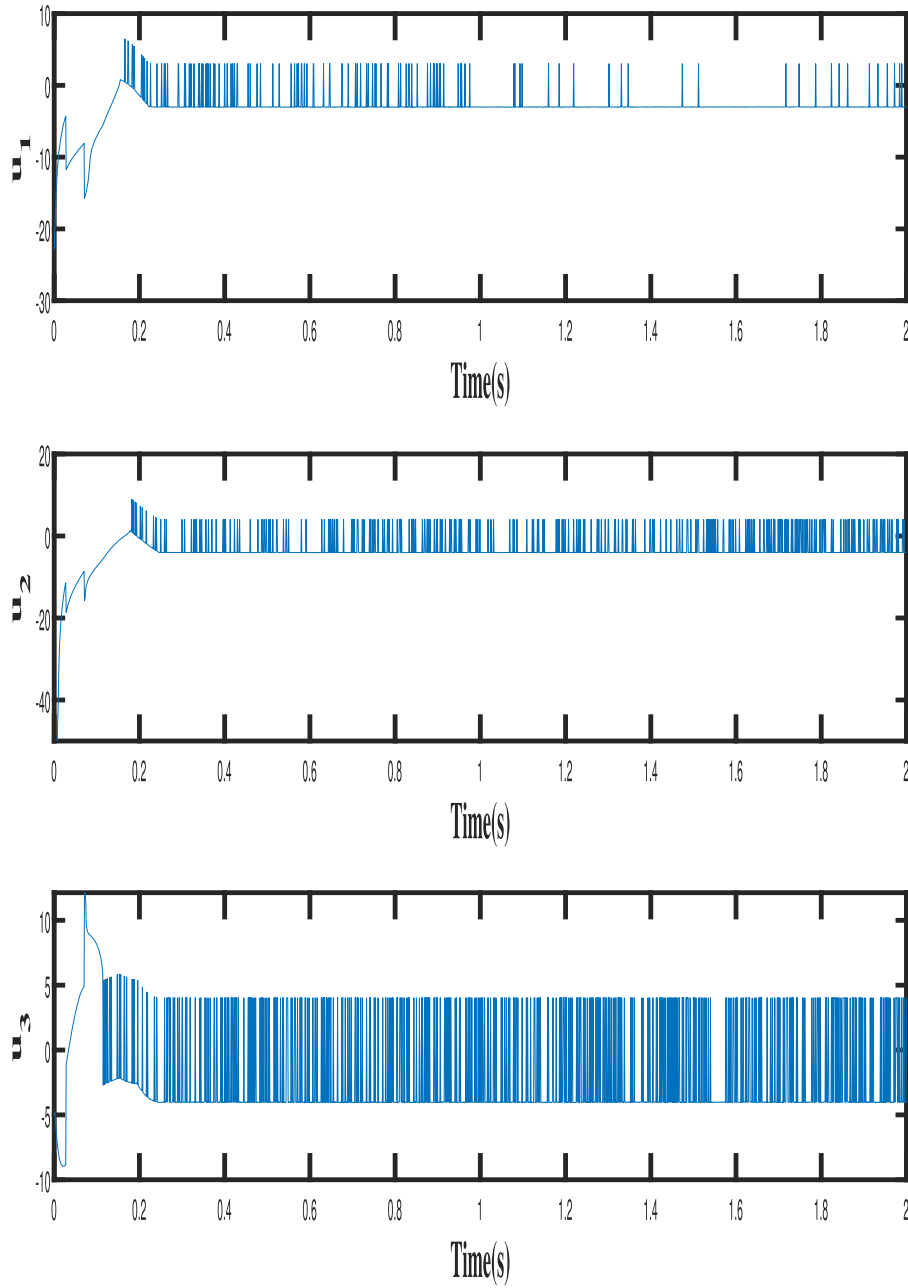
Consider the master system, at transmitter side

$$\begin{cases} \dot{x}_1 = -ax_1 + x_2x_3 + x_2 \\ \dot{x}_2 = cx_2 - x_1x_3 + dx_3 - x_4 \\ \dot{x}_3 = bx_1x_2 - hx_3 + x_2x_4 \\ \dot{x}_4 = x_2 \end{cases} \quad (39)$$

Consider the slave system, at receiver side

$$\begin{cases} \dot{y}_1 = -ay_1 + y_2y_3 + y_2 \\ \dot{y}_2 = cy_2 - y_1y_3 + dy_3 - y_4 \\ \dot{y}_3 = by_1y_2 - hy_3 + y_2y_4 \\ \dot{y}_4 = y_2 \end{cases} \quad (40)$$

Now, we choose the fourth state variable x_4 of master system as a driving variable to drive the slave system, then slave system (40) can be

Fig. 15. Controllers u_1, u_2 and u_3 .

rewritten as follows,

$$\begin{cases} \dot{y}_1 = -ay_1 + y_2y_3 + y_2 + u_1 \\ \dot{y}_2 = cy_2 - y_1y_3 + dy_3 - x_4 + u_2 \\ \dot{y}_3 = by_1y_2 - hy_3 + y_2x_4 + u_3 \end{cases} \quad (41)$$

where u_1, u_2, u_3 are the controllers. The error system between master system(39) and slave system (41) is defined as: $e_1 = y_1 - x_1, e_2 = y_2 - x_2$ and $e_3 = y_3 - x_3$.

Hence, the dynamics of error system is,

$$\begin{cases} \dot{e}_1 = -ae_1 + e_2 + y_2e_3 + x_3e_2 + \Delta e_1 + u_1 \\ \dot{e}_2 = ce_2 + de_3 - y_1e_3 - x_3e_1 + \Delta e_2 + u_2 \\ \dot{e}_3 = by_1e_2 + bx_2e_1 - he_3 + x_4e_2 + \Delta e_3 + u_3 \end{cases} \quad (42)$$

where $\Delta e_1, \Delta e_2, \Delta e_3$ are represent the disturbance due to channel noise.

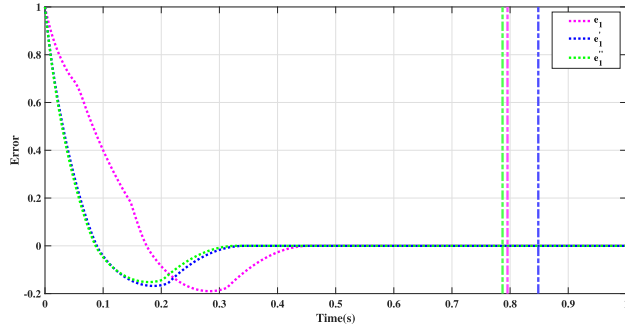
Assumption (A1) :- Let there exist positive constants Δ_1, Δ_2 and Δ_3 such that

$$|\Delta e_1| < \Delta_1, |\Delta e_2| < \Delta_2, |\Delta e_3| < \Delta_3. \quad (43)$$

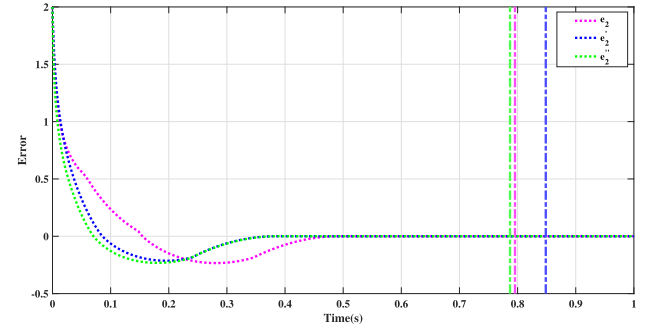
Remark 11. It is noted that during the transmission of encrypted signals over a public channel, disturbances may arise due to channel noise. In this article we talk about only bounded disturbance (either vanishing or non-vanishing). To reject these disturbances and achieve reliable synchronization, a sliding mode control strategy has been designed. In this regard, we now proceed to design NSTSMC along with the corresponding controller.

4.3. Non-singular terminal sliding surface and controller design

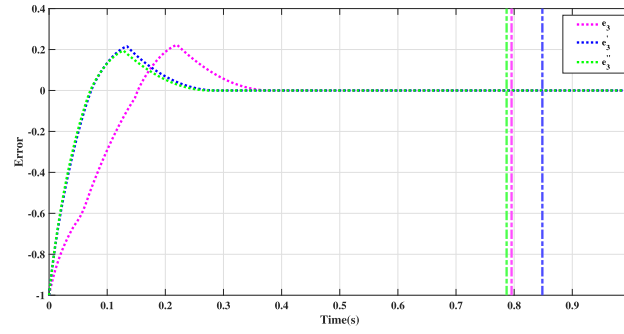
To guarantee that the origin of the system(42) is fixed time stable under the influence of unknown but bounded disturbances ($\Delta e_1, \Delta e_2$



(a)

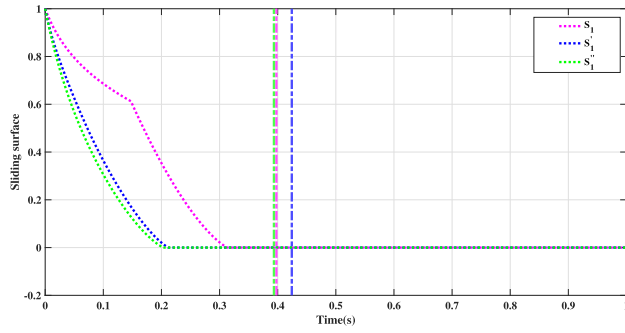


(b)

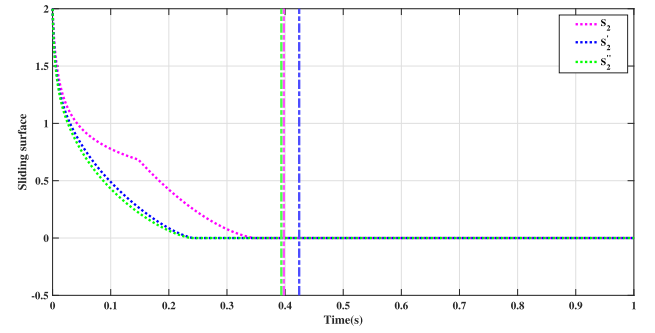


(c)

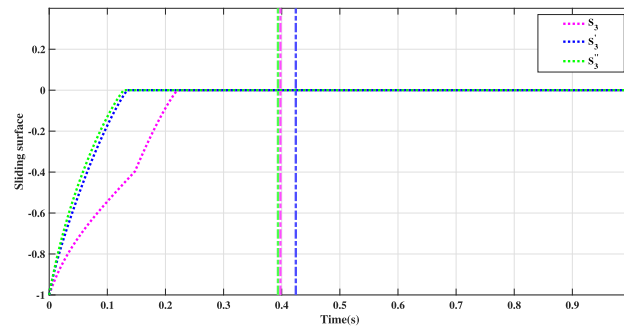
Fig. 16. State trajectories of the error system with initial condition (1,2,-1) under proposed control method (magenta curve), control method in Li et al. (2019) (blue curve), control method in Li et al. (2020) (green curve).



(a)



(b)



(c)

Fig. 17. Sliding surfaces of (44),(63), and (65).

and Δe_3), we design the sliding surface and corresponding controller as follows:

$$S_j = e_j + \int_0^t \left(-\frac{a_1}{2} e_j + \frac{b_1}{2} |e_j|^{\mu_1+2v_1} \cdot \text{sign}(e_j) \right) d\tau, j = 1, 2, 3 \quad (44)$$

$$\begin{cases} u_1 = ae_1 - e_2 - y_2 e_3 - x_3 e_2 + \frac{a_1}{2} e_1 - \frac{b_1}{2} |e_1|^{\mu_1+2v_1} \cdot \text{sign}(e_1) \\ \quad + \frac{a_2}{2} S_1 - \frac{b_2}{2} |S_1|^{\mu_2+2v_2} \cdot \text{sign}(S_1) - \Delta_1 \text{sign}(S_1) \\ u_2 = -ce_2 - de_3 + y_1 e_3 + x_3 e_1 + \frac{a_1}{2} e_2 - \frac{b_1}{2} |e_2|^{\mu_1+2v_1} \cdot \text{sign}(e_2) \\ \quad + \frac{a_2}{2} S_2 - \frac{b_2}{2} |S_2|^{\mu_2+2v_2} \cdot \text{sign}(S_2) - \Delta_2 \text{sign}(S_2) \\ u_3 = -by_1 e_2 - bx_2 e_1 + he_3 - x_4 e_2 + \frac{a_1}{2} e_3 - \frac{b_1}{2} |e_3|^{\mu_1+2v_1} \cdot \text{sign}(e_3) \\ \quad + \frac{a_2}{2} S_3 - \frac{b_2}{2} |S_3|^{\mu_2+2v_2} \cdot \text{sign}(S_3) - \Delta_3 \text{sign}(S_3) \end{cases} \quad (45)$$

where $b_i > 0, a_i < b_i, 0 \leq \mu_i < 1 < v_i$, for $i = 1, 2$.

4.4. Synchronization of master system and slave system

Theorem 4. Once the trajectories of system (42) reaches on sliding surface (44), then the origin $e_i(t) = 0, \forall i = 1, 2, 3$ of system (42) is $FX_T S$. That is, master and slave system are synchronized in fixed time $T_{\max}^6$.

Where

$$T_{\max}^6 = \begin{cases} \frac{2}{b_1(1-\mu_1)} + \frac{2.3}{b_1(\mu_1+2v_1-1)} & ; a_1 = 0 \\ \frac{2}{a_1(1-\mu_1)} \ln \frac{b_1}{b_1-a_1} + \frac{2}{a_1(\mu_1+2v_1-1)} \ln \frac{b_1 \cdot 3^{\frac{1-\mu_1-2v_1}{2}}}{b_1 \cdot 3^{\frac{1-\mu_1-2v_1}{2}} - a_1} & ; a_1 \neq 0 \end{cases}, \quad (46)$$

and $b_i > 0, 0 \leq \mu_i < 1 < v_i$ and $a_i < \min\{b_i, b_i \cdot 3^{\frac{1-\mu_i-2v_i}{2}}\}$.

Proof. When the trajectories of system (42) reaches on sliding surface (44), that is $S_j = 0$, then $\dot{S}_j = 0, j = 1, 2, 3$.

Therefore from Eq. (44), one can easily get

$$\dot{e}_j = \frac{a_1}{2} e_j - \frac{b_1}{2} |e_j|^{\mu_1+2v_1} \cdot \text{sign}(e_j), j = 1, 2, 3 \quad (47)$$

Now, let us consider the Lyapunov function $\mathcal{V}_1 = \sum_{j=1}^3 e_j^2$, then the time derivative of $\mathcal{V}_1$ along the trajectories of (47) is obtained as

$$\begin{aligned} \dot{\mathcal{V}}_1 &= 2 \sum_{j=1}^3 e_j \dot{e}_j \\ &= 2 \sum_{j=1}^3 e_j \left(\frac{a_1}{2} e_j - \frac{b_1}{2} |e_j|^{\mu_1+2v_1} \cdot \text{sign}(e_j) \right) \\ &= a_1 \sum_{j=1}^3 e_j^2 - b_1 \sum_{j=1}^3 |e_j|^{\mu_1+2v_1} \cdot \text{sign}(e_j) \cdot e_j \\ &= a_1 \mathcal{V}_1 - b_1 \sum_{j=1}^3 (e_j^2)^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}}. \end{aligned} \quad (48)$$

From Lemma (4), one can easily obtain the following inequalities:

1. If $0 < \mathcal{V}_1 < 1, 0 < \frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2} = \frac{\mu_1+1}{2} < 1$, then using Lemma (4) and from (48), one can obtain

$$\begin{aligned} \dot{\mathcal{V}}_1 &= a_1 \mathcal{V}_1 - b_1 \sum_{j=1}^3 (e_j^2)^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}} \\ &\leq a_1 \mathcal{V}_1 - b_1 \left(\sum_{j=1}^3 e_j^2 \right)^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}} \\ &= a_1 \mathcal{V}_1 - b_1 \mathcal{V}_1^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}}. \end{aligned} \quad (49)$$

2. If $\mathcal{V}_1 \geq 1, \frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2} = \frac{\mu_1+1}{2} + v_1 > 1$, then using Lemma (4) and from (48), one can obtain

$$\dot{\mathcal{V}}_1 = a_1 \mathcal{V}_1 - b_1 \sum_{j=1}^3 (e_j^2)^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}}$$

$$\begin{aligned} &\leq a_1 \mathcal{V}_1 - b_1 \cdot 3^{1-(\frac{\mu_1+1}{2}+v_1)} \left(\sum_{j=1}^3 e_j^2 \right)^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}} \\ &= a_1 \mathcal{V}_1 - b_1 \cdot 3^{1-(\frac{\mu_1+1}{2}+v_1)} \mathcal{V}_1^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}}. \end{aligned} \quad (50)$$

Therefore, from (49) and (50), it follows that

$$\dot{\mathcal{V}}_1 \leq \begin{cases} a_1 \mathcal{V}_1 - b_1 \mathcal{V}_1^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}} & ; 0 < \mathcal{V}_1 < 1 \\ a_1 \mathcal{V}_1 - b_1 \cdot 3^{1-(\frac{\mu_1+1}{2}+v_1)} \mathcal{V}_1^{\frac{\mu_1+2v_1 \cdot \mathcal{H}(\mathcal{V}_1-1)+1}{2}} & ; \mathcal{V}_1 \geq 1 \end{cases}. \quad (51)$$

From Theorem 2 and Eq. (51), the error system (42) will reach to zero in fixed time and the estimated upper bound of settling time is given by,

$$T(X_0) \leq T_{\max}^6 = \begin{cases} \frac{2}{b_1(1-\mu_1)} + \frac{2.3}{b_1(\mu_1+2v_1-1)} & ; a_1 = 0 \\ \frac{2}{a_1(1-\mu_1)} \ln \frac{b_1}{b_1-a_1} + \frac{2}{a_1(\mu_1+2v_1-1)} \ln \frac{b_1 \cdot 3^{\frac{1-\mu_1-2v_1}{2}}}{b_1 \cdot 3^{\frac{1-\mu_1-2v_1}{2}} - a_1} & ; a_1 \neq 0 \end{cases}.$$

The mathematical proof of Theorem 4 has been completed. $\square$

Theorem 5. If assumption (A1) is hold, then the error system (42) will reach to the sliding surface (44) in fixed time under the controller (45) with parameters a_2, b_2, μ_2, v_2 such that $b_2 > 0, 0 \leq \mu_2 < 1 < v_2$ and $a_2 < \min\{b_2, b_2 \cdot 3^{\frac{1-\mu_2-2v_2}{2}}\}$. Moreover, the estimated upper bound of settling time is,

$$T(X_0) \leq T_{\max}^7 = \begin{cases} \frac{2}{b_2(1-\mu_2)} + \frac{2.3}{b_2(\mu_2+2v_2-1)} & ; a_2 = 0 \\ \frac{2}{a_2(1-\mu_2)} \ln \frac{b_2}{b_2-a_2} + \frac{2}{a_2(\mu_2+2v_2-1)} \ln \frac{b_2 \cdot 3^{\frac{1-\mu_2-2v_2}{2}}}{b_2 \cdot 3^{\frac{1-\mu_2-2v_2}{2}} - a_2} & ; a_2 \neq 0 \end{cases}. \quad (52)$$

Proof. From (44),

$$S_j = e_j + \int_0^t \left(-\frac{a_1}{2} e_j + \frac{b_1}{2} |e_j|^{\mu_1+2v_1} \cdot \text{sign}(e_j) \right) d\tau, j = 1, 2, 3 \quad (53)$$

Differentiate (53) with respect to t , one can obtain

$$\dot{S}_j = \dot{e}_j + \left(-\frac{a_1}{2} e_j + \frac{b_1}{2} |e_j|^{\mu_1+2v_1} \cdot \text{sign}(e_j) \right). \quad (54)$$

Consider the Lyapunov function $\mathcal{V}_2 = \sum_{j=1}^3 S_j^2$, then the time derivative of $\mathcal{V}_2$ along the trajectories of (54) is obtained as

$$\begin{aligned} \dot{\mathcal{V}}_2 &= 2 \sum_{j=1}^3 S_j \dot{S}_j \\ &= 2 \sum_{j=1}^3 S_j \left(\dot{e}_j + \left(-\frac{a_1}{2} e_j + \frac{b_1}{2} |e_j|^{\mu_1+2v_1} \cdot \text{sign}(e_j) \right) \right). \end{aligned} \quad (55)$$

After substituting the values of $\dot{e}_j$ from (42) and using the controllers u_j from (45), one can easily calculate from (55) that,

$$\begin{aligned} \dot{\mathcal{V}}_2 &= 2 \sum_{j=1}^3 S_j \left(\frac{a_2}{2} S_j - \frac{b_2}{2} |S_j|^{\mu_2+2v_2} \cdot \text{sign}(S_j) - \Delta_j \text{sign}(S_j) + \Delta e_j \right) \\ &= a_2 \sum_{j=1}^3 S_j^2 - b_2 \sum_{j=1}^3 |S_j|^{\mu_2+2v_2} \cdot \text{sign}(S_j) \cdot S_j - \Delta_j S_j + S_j \Delta e_j \\ &\leq a_2 \mathcal{V}_2 - b_2 \sum_{j=1}^3 (S_j^2)^{\frac{\mu_2+2v_2 \cdot \mathcal{H}(\mathcal{V}_2-1)+1}{2}} - (\Delta_j - |\Delta e_j|) |S_j|. \end{aligned} \quad (56)$$

Using assumption (A1), from (56) one can obtain,

$$\dot{\mathcal{V}}_2 \leq a_2 \mathcal{V}_2 - b_2 \sum_{j=1}^3 (S_j^2)^{\frac{\mu_2+2v_2 \cdot \mathcal{H}(\mathcal{V}_2-1)+1}{2}}. \quad (57)$$

Now, from Lemma (4), one can easily obtain the following inequalities:

1. If $0 < \mathcal{V}_2 < 1$, $0 < \frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2} = \frac{\mu_2+1}{2} < 1$, then using Lemma (4) and from (57), one can obtain that

$$\begin{aligned} \dot{\mathcal{V}}_2 &\leq a_2 \mathcal{V}_2 - b_2 \sum_{j=1}^3 (S_j^2)^{\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2}} \\ &\leq a_2 \mathcal{V}_2 - b_2 \left(\sum_{j=1}^3 S_j^2 \right)^{\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2}} \\ &\leq a_2 \mathcal{V}_2 - b_2 \mathcal{V}_2^{\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2}}. \end{aligned} \quad (58)$$

2. If $\mathcal{V}_2 \geq 1$, $\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2} = \frac{\mu_2+1}{2} + \nu_2 > 1$, then using Lemma (4) and from (57), one can obtain that

$$\begin{aligned} \dot{\mathcal{V}}_2 &\leq a_2 \mathcal{V}_2 - b_2 \sum_{j=1}^3 (S_j^2)^{\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2}} \\ &\leq a_2 \mathcal{V}_2 - b_2 \cdot 3^{1-(\frac{\mu_2+1}{2}+\nu_2)} \left(\sum_{j=1}^3 S_j^2 \right)^{\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2}} \\ &\leq a_2 \mathcal{V}_2 - b_2 \cdot 3^{1-(\frac{\mu_2+1}{2}+\nu_2)} \mathcal{V}_2^{\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2}}. \end{aligned} \quad (59)$$

Therefore, from (58) and (59), it follows that

$$\dot{\mathcal{V}}_2 \leq \begin{cases} a_2 \mathcal{V}_2 - b_2 \mathcal{V}_2^{\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2}} & ; 0 < \mathcal{V}_2 < 1 \\ a_2 \mathcal{V}_2 - b_2 \cdot 3^{1-(\frac{\mu_2+1}{2}+\nu_2)} \mathcal{V}_2^{\frac{\mu_2+2\nu_2.H(\mathcal{V}_2-1)+1}{2}} & ; \mathcal{V}_2 \geq 1 \end{cases}. \quad (60)$$

From Theorem 2 and Eq. (60), sliding surface (44) will reach zero in fixed time and the estimated upper bound of settling time is given by,

$$T(X_0) \leq T_{\max}^7 = \begin{cases} \frac{2}{b_2(1-\mu_2)} + \frac{2 \cdot 3^{\frac{(\mu_2+2\nu_2-1)}{2}}}{b_2(\mu_2+2\nu_2-1)} & ; a_2 = 0 \\ \frac{2}{a_2(1-\mu_2)} \ln \frac{b_2}{b_2-a_2} + \frac{2}{a_2(\mu_2+2\nu_2-1)} \ln \frac{b_2 \cdot 3^{\frac{1-\mu_2-2\nu_2}{2}}}{b_2 \cdot 3^{\frac{1-\mu_2-2\nu_2}{2}} - a_2} & ; a_2 \neq 0 \end{cases}.$$

The proof of Theorem 5 has been completed. $\square$

Thus, through the combination of Theorems 4 and 5, we conclude that the error trajectories of the system (42) are $FX_T.S$. $T = T_{\max}^6 + T_{\max}^7$ any for initial condition.

4.5. Encryption and decryption process

In this section, we considered the encryption-decryption processes proposed in Benkouider et al. (2020), which is more complex and offers a higher level of security as compared to many existing encryption schemes presented in previous works.

4.5.1. Encryption

For encryption of plain message $m(t)$, we use three states x_1, x_2 and x_3 of master system and for enhancing the level of security, we divide encryption process into two cases based on the sign of x_1 . Thus the encrypted message is obtained by the following process.

If $x_1 > 0$, then encrypted message is,

$$E_m(t) = l_1 m + l_2 x_1 + l_3 x_2.$$

If $x_1 \leq 0$, then encrypted message is,

$$E_m(t) = r_1 m + r_2 x_1 + r_3 x_3.$$

Where m is plain message, l_1, l_2, l_3, r_1, r_2 and r_3 are the encryption parameters or secret keys that are used to increase the complexity and level of security.

4.5.2. Decryption

The retrieved message can be obtained after the master and slave systems have been synchronized, as seen below.

If $y_1 > 0$, then decrypted message is,

$$D_m(t) = \frac{1}{l_1} [E_m(t) - l_2 y_1 - l_3 y_2].$$

If $y_1 \leq 0$, then decrypted message is,

$$D_m(t) = \frac{1}{r_1} [E_m(t) - r_2 y_1 - r_3 y_3].$$

The parameters l_1, l_2, l_3, r_1, r_2 and r_3 are the same as in the encryption procedure.

Remark 12. In secure communication, to prevent data loss or distortion of the transmitted message, it is essential to effectively reject disturbances and uncertainties. This requirement is particularly critical in chaotic secure communication systems, where the master and slave systems are employed for signal encryption and decryption. Since chaotic systems are inherently unpredictable and unstable, synchronization between the master and slave systems is the key to ensuring both reliability and security. In order to address this issue and achieve robust synchronization in the presence of channel noise, a sliding mode control scheme is developed in the present work. The proposed SMC not only guarantees that the synchronization error converges to zero in fixed time but also provides strong disturbance rejection capability. As a result, the proposed method enables accurate message recovery at the receiver end while enhancing the robustness and security of the communication scheme.

5. Numerical simulations

This section includes some simulations to validate the results established in previous remarks and theorems.

Simulation 1

The effectiveness of the conditions $a < b$ described in Theorem 1 is examined in this simulation. For this let us consider the non-linear scalar system

$$\dot{X} = aX - b|X|^{\frac{1}{2} + \frac{3}{2}H(|X|-1)} \text{sign}(X). \quad (61)$$

Consider the Lyapunov function $\mathcal{V}(X) = |X|$, then the time derivative of $\mathcal{V}$ along the trajectories of (61) is,

$$\dot{\mathcal{V}}(X) = a\mathcal{V}(X) - b(\mathcal{V}(X))^{\frac{1}{2} + \frac{3}{2}H(\mathcal{V}-1)} \quad (62)$$

Here $\mu = \frac{1}{2}$ and $\nu = \frac{3}{2}$.

Let consider two case first satisfying $a < b$ and second violating $a < b$.

Case 1. $a = 3$ and $b = 5$.

Case 2. $a = 6$ and $b = 1$.

From Fig. 6, it can be seen that if the parameters satisfy the condition $a < b$, then the trajectories of (61) converge to zero, whereas if the condition $a < b$ is violated, they may not converge to zero.

Simulation 2

This simulation demonstrates that the states of Eq. (42) converge to the sliding surface defined by Eq. (44) within a fixed time $T_{\max}^7$. The following parameters are selected for the simulation: $a_2 = -1, b_2 = 23, \mu_2 = 0.5$ and $\nu_2 = 2$. Within the fixed time $T_{\max}^7 = 0.3190$, the sliding surfaces reach zero, as seen in Fig. 7.

Simulation 3

This section presents a numerical verification of the synchronization between the master and slave systems, as described in Eq. (39) and (40). For this consider the values of parameters of master system (39) and slave system (40) as $a = 4, b = 2, c = 5.5, d = 10$ and $h = 19$ so that system exhibits a hyper-chaotic behavior. Also consider the initial conditions of master system (39) as $x_1(0) = 1, x_2(0) = 2, x_3(0) = 1$ and $x_4(0) = 1$.

The initial conditions of slave system (40) as $y_1(0) = 2, y_2(0) = 4, y_3(0) = 0$ and $y_4(0) = 2$ and the disturbances $\triangle e_1, \triangle e_2, \triangle e_3$ and their bounds $\triangle_1, \triangle_2, \triangle_3$ considered as $2\cos(t), 2 + \sin(t), \cos(t) + \sin(t)$ and $3, 3, 2$ respectively. To reduce computational complexity $a_1 = a_2, b_1 = b_2, \mu_1 = \mu_2, \nu_1 = \nu_2$.

Now, we examine three cases for the parameter $a_1 = a_2$ as follows.

Case 1. $a_1 = a_2 = -1$ and other parameters as $b_1 = 23, \mu_1 = 0.5, \nu_1 = 2$, then upper bound of settling time is obtained by $T = T_{\max}^6 + T_{\max}^7 = 0.3190 + 0.3190 = 0.6380$. Fig. 8 Shows the synchronization achieved between the master-slave systems, as described in Eqs. (39) and (40), respectively. Simulation results confirm that all states synchronized within a fixed time $T = 0.6380$.

Case 2. $a_1 = a_2 = 0$ and other parameters as $b_1 = 30, \mu_1 = 0.5, \nu_1 = 3$, then upper bound of settling time is obtained by $T = T_{\max}^6 + T_{\max}^7 = 0.3820 + 0.3820 = 0.7640$. Fig. 9 illustrates the synchronization between the master system and the slave system, as described in Eqs. (39) and (40), respectively. It can be observed that synchronization of all states is effectively achieved within a fixed time $T = 0.7640$.

Case 3. $a_1 = a_2 = 2$ and other parameters as $b_1 = 29, \mu_1 = 0.5, \nu_1 = 2$, then upper bound of settling time is obtained by $T = T_{\max}^6 + T_{\max}^7 = 0.3252 + 0.3252 = 0.6504$. Fig. 10 depicts the synchronization between the master system (39) and the slave system (40). The synchronization of all states is successfully attained within a fixed time $T = 0.6504$, as observed.

Simulation 4

We will demonstrate the performance of the encryption procedure. Subsequently, after synchronization is achieved, the decryption scheme will be executed. we choose the encryption parameters as $l_1 = 1, l_2 = 1, l_3 = 1, r_1 = 1, r_2 = 1, r_3 = 1$.

Fig. 11 shows the complex chaotic behavior of state trajectories x_1, x_2 and x_3 of equation (39). Now we choose $m(t) = \sin(100\pi t)$ as a plain message depicted in Fig. 12. Also Figs. 13 and 14, shows the encrypted message and decrypted message respectively. In this simulation, we will be considering the initial conditions and parameters values same as those in case 1 of simulation 3. As a result, we can see that after small time $T = 0.6380$ the obtained message is same as the original message. That is, the plain message is successfully decrypted and retrieved at the receiving end within a short time.

Simulation 5 The time evolution of controllers defined in Eq. (45) are depicted in Fig. 15.

Simulation 6 Since accurate estimation of the settling time is an important criterion. Therefore, in this simulation, different types of sliding surfaces and corresponding controllers based on existing results are considered and compared with the proposed sliding surface in terms of accuracy.

Based on Lemma 2, fixed-time control is achieved using the following sliding-mode surface and controller

$$S'_j = e'_j + \int_0^t \left(-\frac{a_1}{2} e'_j + \frac{b_1}{2} |e'_j|^{\mu_1} \cdot \text{sign}(e'_j) + \frac{c_1}{2} |e'_j|^{2\nu_1} \cdot \text{sign}(e'_j) \right) d\tau, \quad (63)$$

$$u'_j = F'_j + \frac{a_1}{2} e'_j - \frac{b_1}{2} |e'_j|^{\mu_1} \cdot \text{sign}(e'_j) - \frac{c_1}{2} |e'_j|^{2\nu_1} \cdot \text{sign}(e'_j) + \frac{a_2}{2} S'_j - \frac{b_2}{2} |S'_j|^{\mu_2} \cdot \text{sign}(S'_j) - \frac{c_2}{2} |S'_j|^{2\nu_2} \cdot \text{sign}(S'_j) - \triangle_j \text{sign}(S'_j). \quad (64)$$

Similarly, based on Lemma 3, fixed-time control is achieved using the following sliding-mode surface and controller

$$S''_j = e''_j + \int_0^t \left(-\frac{a_1}{2} e''_j + \frac{b_1}{2} |e''_j|^{\mu_1} \cdot \text{sign}(e''_j) + \frac{c_1}{2} |e''_j|^{2\nu_1 + \text{sign}(\nu_1 - 1)} \cdot \text{sign}(e''_j) \right) d\tau, \quad (65)$$

$$u''_j = F''_j + \frac{a_1}{2} e''_j - \frac{b_1}{2} |e''_j|^{\mu_1} \cdot \text{sign}(e''_j) - \frac{c_1}{2} |e''_j|^{2\nu_1 + \text{sign}(\nu_1 - 1)} \cdot \text{sign}(e''_j) + \frac{a_2}{2} S''_j$$

$$- \frac{b_2}{2} |S''_j|^{\mu_2} \cdot \text{sign}(S''_j) - \frac{c_2}{2} |S''_j|^{2\nu_2 + \text{sign}(\nu_2 - 1)} \cdot \text{sign}(S''_j) - \triangle_j \text{sign}(S''_j). \quad (66)$$

Where $j = 1, 2, 3$ and F'_j, F''_j are the non-linear functions. To ensure a fair comparison, identical parameter values are adopted as follows: $a_1 = a = 1, b_1 = c_1 = b_2 = c_2 = 12, \mu_1 = \mu_2 = 0.5, \nu_1 = \nu_2 = 2$, and $\nu_1 = |x|$ is employed as the

Lyapunov function. The state trajectories of the error systems are presented in Fig. 16, whereas Fig. 17 illustrates the sliding surfaces defined in Eqs. (44), (63), and (65). Moreover, Figs. 16 and 17 clearly demonstrate that the gap between the true convergence time and its upper bound (magenta curve) is smaller in the proposed SMC compared to the other SMC, thereby confirming the improved accuracy of the proposed approach.

Remark 13. Figs. 16 and 17 present simulation results obtained under identical parameter values and initial conditions. It is easy to obtain from Figs. 16 and 17 that the proposed NSTSS provides a more accurate prediction of the settling-time upper bound compared with the control methods in Li et al. (2019), Li et al. (2020). This is indeed beneficial in practical applications as one generally do not have any idea of the true convergence time and hence, more accurate the prediction more accurate will be the results.

Furthermore, the methods in Li et al. (2019), Li et al. (2020) employ multiple power terms and additional design parameters to enforce convergence across different state regions, resulting in increased tuning complexity and unnecessary control action. In contrast, the proposed formulation adapts the effective dynamics through a Heaviside function, reduces the number of tuning parameters, and applies control effort only when required. Consequently, lower energy consumption is achieved while preserving the desired fixed-time convergence properties.

6. Conclusion

In this paper, Lyapunov-based FX_T -S results for non-linear systems are investigated, some new non-singular-terminal sliding surfaces and corresponding controller have been designed. First, three new theorems on FX_T -S are presented, and the estimated upper bounds of the associated settling time functions are obtained. The accuracy of the proposed result is confirmed through various remarks. Furthermore, as an application, a secure communication scheme has been developed using a chaotic cryptography-based encryption algorithm. In future, we will focus on obtaining similar results for impulsive systems and delayed impulsive systems. Apart from this, efforts will be made to improve the results for predefined and prescribed-time stability.

CRedit authorship contribution statement

Rajesh Kushwah: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation; **Arnab Mapui:** Writing – review & editing, Visualization, Supervision; **Santwana Mukhopadhyay:** Writing – review & editing, Validation, Supervision.

Data availability

The authors declare that all the data generated and analyzed during the study are present in the article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Rajesh Kushwah reports financial support was provided by University Grants Commission. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

One of the authors, Rajesh kushwah, thankfully acknowledges the full financial assistance of the JRF Fellowship (Roll Number MP06010613, Reference number 231610172510) by the University Grant Commission (UGC), India, to carry out the present work.

References

- Afghoul, H., Krim, F., Babes, B., Beddar, A., & Kihel, A. (2018). Design and real time implementation of sliding mode supervised fractional controller for wind energy conversion system under sever working conditions. *Energy Conversion and Management*, 167, 91–101.
- Ahmad, I., Saaban, A. B., Ibrahim, A. B., & Shahzad, M. (2015). A research on active control to synchronize a new 3d chaotic system. *Systems*, 4(1), 2.
- Atassi, A. N., & Khalil, H. K. (2000). Separation results for the stabilization of nonlinear systems using different high-gain observer designs. *Systems & Control Letters*, 39(3), 183–191.
- Babes, B., Boutaghane, A., Hamouda, N., & Mezaache, M. (2019). Design of a robust voltage controller for a DC-DC buck converter using fractional-order terminal sliding mode control strategy. In *2019 international conference on advanced electrical engineering (ICAEE)* (pp. 1–6). IEEE.
- Babes, B., Mekhilef, S., Boutaghane, A., & Rahmani, L. (2021). Fuzzy approximation-based fractional-order nonsingular terminal sliding mode controller for DC-DC buck converters. *IEEE Transactions on Power Electronics*, 37(3), 2749–2760.
- Balta, G., Hisar, C., & Altin, N. (2023). Artificial intelligence based switching frequency regulation with fast terminal sliding mode control for DC-DC step-down converters. *Expert Systems with Applications*, 228, 120331.
- Benkoudier, K., Bouden, T., & Yalcin, M. E. (2020). A snail-shaped chaotic system with large bandwidth: dynamical analysis, synchronization and secure communication scheme. *SN Applied Sciences*, 2, 1–15.
- Bhat, S. P., & Bernstein, D. S. (2000). Finite-time stability of continuous autonomous systems. *SIAM Journal on Control and optimization*, 38(3), 751–766.
- Blasius, B., & Stone, L. (2000). Chaos and phase synchronization in ecological systems. *International Journal of Bifurcation and Chaos*, 10(10), 2361–2380.
- Cao, J., & Wang, L. (2002). Exponential stability and periodic oscillatory solution in BAM networks with delays. *IEEE Transactions on Neural Networks*, 13(2), 457–463.
- Chen, C., Li, L., Peng, H., Yang, Y., Mi, L., & Zhao, H. (2020). A new fixed-time stability theorem and its application to the fixed-time synchronization of neural networks. *Neural networks*, 123, 412–419.
- Chen, X., & Lu, J. (2007). Adaptive synchronization of different chaotic systems with fully unknown parameters. *Physics Letters A*, 364(2), 123–128.
- Corradini, M. L., & Cristofaro, A. (2018). Nonsingular terminal sliding-mode control of nonlinear planar systems with global fixed-time stability guarantees. *Automatica*, 95, 561–565.
- Fu, J., Luo, R., Huang, M., & Su, H. (2021). Fixed time synchronization of a class of chaotic systems based via the saturation control. *Revista mexicana de física*, 67(4).
- Haddad, W. M., Nersesov, S. G., & Du, L. (2008). Finite-time stability for time-varying nonlinear dynamical systems. In *2008 american control conference* (pp. 4135–4139). IEEE.
- Hale, J. K., & Collection, K. M. R. (1980). Ordinary Differential Equations. Pure and applied mathematics. R. E. Krieger. <https://books.google.co.in/books?id=NW1BxwEACAAJ>.
- Hale, J. K., & Lunel, S. M. V. (2003). Stability and control of feedback systems with time delays. *International Journal of Systems Science*, 34(8–9), 497–504.
- Hardy, G. H., Littlewood, J. E., & Pólya, G. (1952). Inequalities. Cambridge university press.
- He, X., Li, X., & Nieto, J. J. (2021). Finite-time stability and stabilization for time-varying systems. *Chaos, Solitons & Fractals*, 148, 111076.
- Hu, C., He, H., & Jiang, H. (2020). Fixed/preassigned-time synchronization of complex networks via improving fixed-time stability. *IEEE Transactions on Cybernetics*, 51(6), 2882–2892.
- Hu, C., Yu, J., Chen, Z., Jiang, H., & Huang, T. (2017). Fixed-time stability of dynamical systems and fixed-time synchronization of coupled discontinuous neural networks. *Neural Networks*, 89, 74–83.
- Istratov, A. A., & Vyvenko, O. F. (1999). Exponential analysis in physical phenomena. *Review of Scientific Instruments*, 70(2), 1233–1257.
- Khalil, H. K. (2002). Nonlinear Systems. Pearson Education. Prentice Hall. https://books.google.co.in/books?id=t_d1QgAACAAJ.
- Kong, F., & Zhu, Q. (2021). Fixed-time stabilization of discontinuous neutral neural networks with proportional delays via new fixed-time stability lemmas. *IEEE Transactions on Neural Networks And Learning Systems*, 34(2), 775–785.
- Kong, F., Zhu, Q., & Huang, T. (2021). New fixed-time stability criteria of time-varying delayed discontinuous systems and application to discontinuous neutral-type neural networks. *IEEE Transactions on Cybernetics*, 53(4), 2358–2367.
- Launay, F., Coirault, P., Cauet, S., & Hutu, F. (2006). Synchronization of two chaotic systems using pid control. *IFAC Proceedings Volumes*, 39(8), 148–153.
- Levant, A. (2013). On fixed and finite time stability in sliding mode control. In *52nd IEEE conference on decision and control* (pp. 4260–4265). IEEE.
- Li, B., Zhang, H., Xiao, B., Wang, C., & Yang, Y. (2022). Fixed-time integral sliding mode control of a high-order nonlinear system. *Nonlinear Dynamics*, 107(1), 909–920.
- Li, N., Wu, X., Feng, J., & Xu, Y. (2019). Fixed-time synchronization in probability of drive-response networks with discontinuous nodes and noise disturbances. *Nonlinear Dynamics*, 97, 297–311.
- Li, N., Wu, X., Feng, J., Xu, Y., & Lü, J. (2020). Fixed-time synchronization of coupled neural networks with discontinuous activation and mismatched parameters. *IEEE Transactions on Neural Networks and Learning Systems*, 32(6), 2470–2482.
- Luo, R., & Zeng, Y. (2015). The adaptive control of unknown chaotic systems with external disturbance via a single input. *Nonlinear Dynamics*, 80(1), 989–998.
- Mapui, A., Jamal, M. A., & Mukhopadhyay, S. (2025). On optimal upper bound for the settling time of fixed-time stable systems and its application in secure communication. *Nonlinear Dynamics*, 113(6), 5359–5381.
- Moulay, E., & Perruquetti, W. (2006). Finite time stability and stabilization of a class of continuous systems. *Journal of Mathematical analysis and applications*, 323(2), 1430–1443.
- Moulay, E., & Perruquetti, W. (2008). Finite time stability conditions for non-autonomous continuous systems. *International Journal of control*, 81(5), 797–803.
- Moysis, L., Petavratzis, E., Marwan, M., Volos, C., Nistazakis, H., & Ahmad, S. (2020). Analysis, synchronization, and robotic application of a modified hyperjerk chaotic system. *Complexity*, 2020(1), 2826850.
- Mu, C., & He, H. (2017). Dynamic behavior of terminal sliding mode control. *IEEE Transactions on Industrial Electronics*, 65(4), 3480–3490.
- Nestor, T., Belazi, A., Abd-El-Atty, B., Aslam, M. N., Volos, C., De Dieu, N. J., & Abd El-Latif, A. (2022). A new 4d hyperchaotic system with dynamics analysis, synchronization, and application to image encryption. *Symmetry*, 14(2), 424.
- Ott, E., Grebogi, C., & Yorke, J. A. (1990). Controlling chaos. *Physical review letters*, 64(11), 1196.
- Pan, H., Zhang, G., Ouyang, H., & Mei, L. (2020). A novel global fast terminal sliding mode control scheme for second-order systems. *IEEE Access*, 8, 22758–22769.
- Pang, S., Feng, Y., & Liu, Y. (2016). Finite-time synchronization of chaotic systems with different dimension and secure communication. *Mathematical Problems in Engineering*, 2016(1), 7693547.
- Parsegov, S., Polyakov, A., & Shcherbakov, P. (2012). Nonlinear fixed-time control protocol for uniform allocation of agents on a segment. In *2012 IEEE 51st IEEE conference on decision and control (CDC)* (pp. 7732–7737). IEEE.
- Polyakov, A. (2011). Nonlinear feedback design for fixed-time stabilization of linear control systems. *IEEE Transactions On Automatic Control*, 57(8), 2106–2110.
- Rabiee, H., Ataei, M., & Ekramian, M. (2019). Continuous nonsingular terminal sliding mode control based on adaptive sliding mode disturbance observer for uncertain nonlinear systems. *Automatica*, 109, 108515.
- Su, H., Luo, R., Huang, M., & Fu, J. (2023). Fast convergence control of a class of uncertain chaotic systems with input nonlinearity by using a new sliding mode controller. *European Journal of Control*, 69, 100751.
- Sun, Y., Wang, F., Liu, Z., Zhang, Y., & Chen, C. L. P. (2020). Fixed-time fuzzy control for a class of nonlinear systems. *IEEE Transactions on Cybernetics*, 52(5), 3880–3887.
- Tran, M.-D., & Kang, H.-J. (2015). Nonsingular terminal sliding mode control of uncertain second-order nonlinear systems. *Mathematical Problems in Engineering*, 2015(1), 181737.
- Utkin, V., Poznyak, A., Orlov, Y., & Polyakov, A. (2020). Conventional and high order sliding mode control. *Journal of the Franklin Institute*, 357(15), 10244–10261.
- Utkin, V. I. (2013). Sliding modes in control and optimization. Springer Science & Business Media.
- Vaidyanathan, S., & Azar, A. T. (2021). An introduction to backstepping control. In *Backstepping control of nonlinear dynamical systems* (pp. 1–32). Elsevier.
- Wan, Z., Wang, C., Luo, X., Lin, Y., & Huang, T. (2014). Generating variable number of wings from a novel four-dimensional hyperchaotic system with one equilibrium. *Optik*, 125(3), 1371–1376.
- Wang, X., Tong, D., Chen, Q., & Zhou, W. (2024). Fixed-time synchronization and energy consumption prediction of interconnected memristive neural networks with discontinuous activation functions. *Circuits, Systems, and Signal Processing*, 43(11), 6977–6995.
- Wu, Y., Yu, X., & Man, Z. (1998). Terminal sliding mode control design for uncertain dynamic systems. *Systems & Control Letters*, 34(5), 281–287.
- Xiao, Q., Liu, H., & Wang, Y. (2021). An improved finite-time and fixed-time stable synchronization of coupled discontinuous neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 34(7), 3516–3526.
- Xu, C., Jiang, M., & Hu, J. (2023). Fixed-time synchronization of complex-valued memristive competitive neural networks based on two novel fixed-time stability theorems. *Neural Computing and Applications*, 35(30), 22605–22620.
- Yau, H.-T., Chen, C.-K., & Chen, C.-L. (2000). Sliding mode control of chaotic systems with uncertainties. *International Journal of Bifurcation and Chaos*, 10(05), 1139–1147.
- Yu, S., Yu, X., Shirinzadeh, B., & Man, Z. (2005). Continuous finite-time control for robotic manipulators with terminal sliding mode. *Automatica*, 41(11), 1957–1964.
- Yu, X., & Zhihong, M. (2002). Fast terminal sliding-mode control design for nonlinear dynamical systems. *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, 49(2), 261–264.
- Zhang, J., Yang, J., Gan, Q., Wu, H., & Cao, J. (2024). Improved fixed-time stability analysis and applications to synchronization of discontinuous complex-valued fuzzy cellular neural networks. *Neural Networks*, 179, 106585.
- Zhihong, M., Paplinski, A. P., & Wu, H. R. (1994). A robust MIMO terminal sliding mode control scheme for rigid robotic manipulators. *IEEE Transactions On Automatic Control*, 39(12), 2464–2469.
- Zuo, Z. (2015). Non-singular fixed-time terminal sliding mode control of non-linear systems. *IET Control Theory & Applications*, 9(4), 545–552.